

SQP Electricity — master question bank — current verified release

Current published verified bank only; no expanded v2 audit claimed

Official-source question archive | 48 source occurrences

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Source Q7 | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 3

cbse_sqp_2017-18_electricity_q7 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

	(a) Lights and fans : (b) Appliances of 2kW or more power :	
7	Draw a circuit diagram of an electric circuit containing a cell, a key , an ammeter , a resistor of 4Ω in series with a combination of two resistors (8Ω each) in parallel and a voltmeter across parallel combination. Each of them dissipate maximum energy and can withstand a maximum power of 16W without melting. Find the maximum current that can flow through the three resistors.	3
8	In the electrolysis of water	2

Source Q26 | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 2

cbse_sqp_2017-18_electricity_q26 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

26	The values of current I flowing in a given resistor for the corresponding values of potential difference V across the resistor are given below:	2

Source Q26 | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 2

cbse_sqp_2017-18_electricity_q26 | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

Plot a graph between V and I and calculate the resistance of the resistor.

Source Q26 OR | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 2

cbse_sqp_2017-18_electricity_q26_OR | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

OR

In a given ammeter, a student sees that needle indicates 17 divisions in ammeter while performing an experiment to verify Ohm's law. If ammeter has 10 divisions between 0 and 0.5A, then what is the value corresponding to 17 divisions?

Source Q12 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 3

cbse_sqp_2018-19_electricity_q12 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

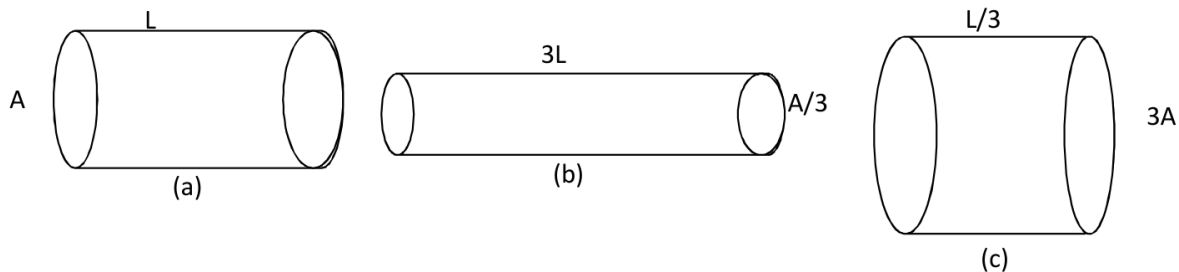
12. Two lamps, one rated 100 W at 220 V and the other 200 W at 220V are connected (i) in series and (ii) in parallel to electric main supply of 220V. Find the current drawn in each case.

3

Source Q13 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 3

cbse_sqp_2018-19_electricity_q13 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

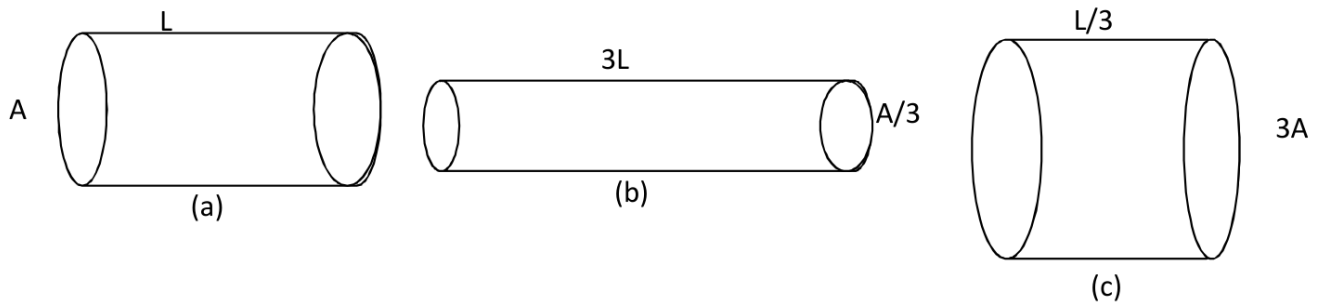
13. The figure below shows three cylindrical copper conductors along with their face areas and lengths. Compare the resistance and the resistivity of the three conductors. Justify your answer. 3



3

Source Q13 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 3

cbse_sqp_2018-19_electricity_q13 | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



Source Q26 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 2

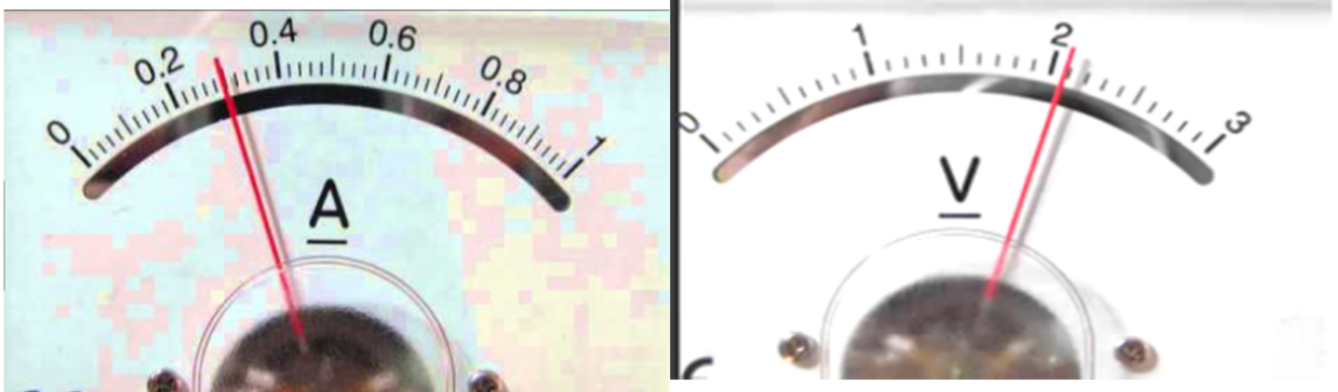
cbse_sqp_2018-19_electricity_q26 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

26. The current flowing through a resistor connected in an electrical circuit and the potential difference developed across its ends are shown in the given ammeter and voltmeter. Find the least count of the voltmeter and ammeter .What is the voltage and the current across the given resistor?

2

Source Q26 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 2

cbse_sqp_2018-19_electricity_q26 | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



Source Q6 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1

cbse_sqp_2019-20_electricity_q6 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

- 6 When a 4V battery is connected across an unknown resistor there is a current of 100 mA in the circuit. The value of the resistance of the resistor is:
- i) $4\ \Omega$
 - ii) $40\ \Omega$
 - iii) $400\ \Omega$
 - iv) $0.4\ \Omega$

Source Q7 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1

cbse_sqp_2019-20_electricity_q7 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

- | | | |
|---|--|---|
| 7 | Unit of electric power may also be expressed as: | 1 |
| | i) volt-ampere | |
| | ii) kilowatt-hour | |
| | iii) watt-second | |
| | iv) joule-second | |

Source Q29 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 5

cbse_sqp_2019-20_electricity_q29 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

- 29 (i) Consider a conductor of resistance 'R', length 'L', thickness 'd' and resistivity ' ρ '. 5
Now this conductor is cut into four equal parts. What will be the new resistivity of each of these parts? Why?
- (ii) Find the resistance if all of these parts are connected in:
- (a) Parallel
 - (b) Series
- (iii) Out of the combinations of resistors mentioned above in the previous part, for a given voltage which combination will consume more power and why?

Source Q14 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1

SQP-2019-20-Q14 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

14 **Assertion:** A fuse wire is always connected in parallel with the mainline. 1

Reason: If a current larger than the specified value flows through the circuit, fuse wire melts.

Source Q14 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1

SQP-2019-20-Q14 | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

For question numbers 13 and 14, two statements are given- one labeled *Assertion (A)* and the other labeled *Reason (R)*. Select the correct answer to these questions from the codes (i), (ii), (iii) and (iv) as given below

- i) Both A and R are true and R is correct explanation of the assertion.
- ii) Both A and R are true but R is not the correct explanation of the assertion.
- iii) A is true but R is false.
- iv) A is false but R is true.

13 **Assertion:** Following are the structural isomers of butane.

1

Source Q9 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 1

sqp_2020-21_q9_main | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

9	Two unequal resistances are connected in parallel. If you are not provided with any other parameters (eg. numerical values of I and R), what can be said about the voltage drop across the two resistors?	1
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Source Q9 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 1

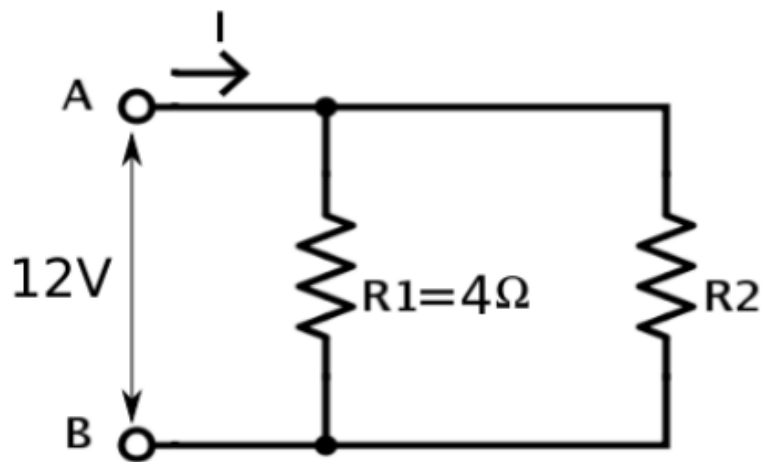
sqp_2020-21_q9_OR | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

OR

Some work is done to move a charge Q from infinity to a point A in space. The potential of the point A is given as V . What is the work done to move this charge from infinity in terms of Q and V ?

26

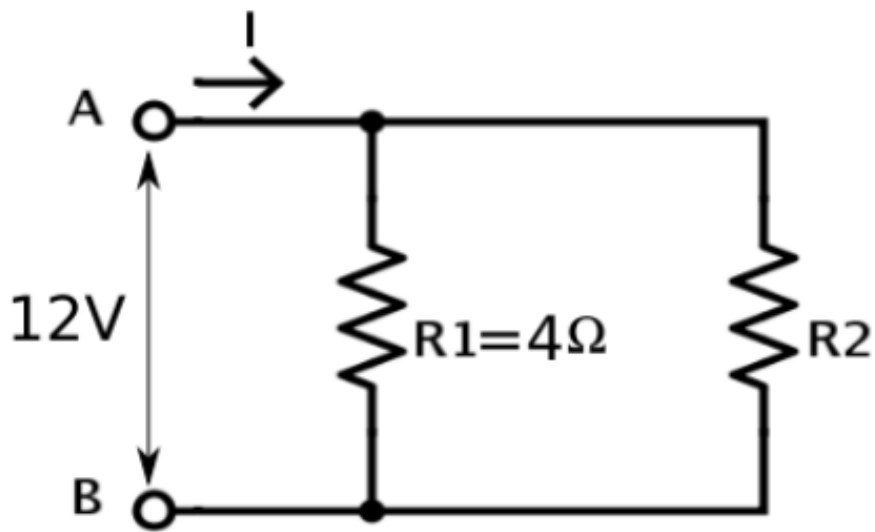
2



A student has two resistors- 2Ω and 3Ω . She has to put one of them in place of R_2 as shown in the circuit. The current that she needs in the entire circuit is exactly 9A. Show by calculation which of the two resistors she should choose.

Source Q26 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 2

sqp_2020-21_q26 | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



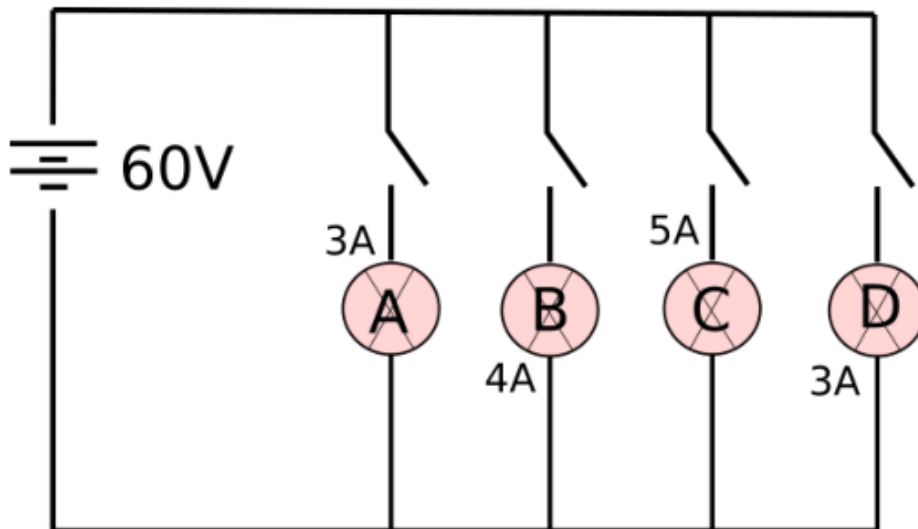
Source Q36 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 5

sqp_2020-21_q36_main | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

36

In the given circuit, A, B, C and D are four lamps connected with a battery of 60V.

5

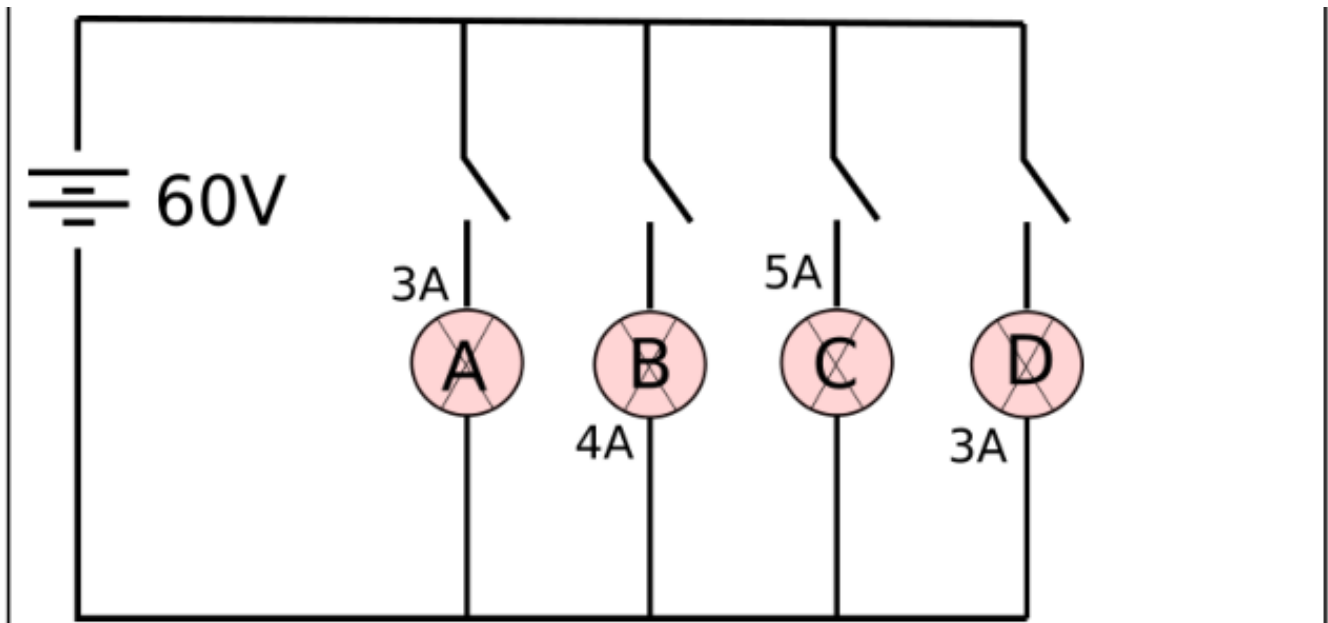


Analyse the circuit to answer the following questions.

- What kind of combination are the lamps arranged in (series or parallel)?
- Explain with reference to your above answer, what are the advantages (any two) of this combination of lamps?
- Explain with proper calculations which lamp glows the brightest?
- Find out the total resistance of the circuit.

Source Q36 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 5

sqp_2020-21_q36_main | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



Source Q11 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

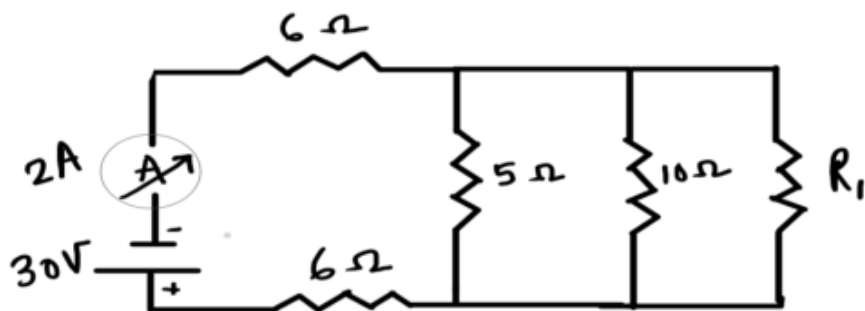
sqp_2021-22_term2_q11 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

11	a. It would cost a man Rs. 3.50 to buy 1.0 kW h of electrical energy from the Main Electricity Board. His generator has a maximum power of 2.0 kW. The generator produces energy at this maximum power for 3 hours. Calculate how much it would cost to buy the same amount of energy from the Main Electricity Board.(1 Mark) b. A student boils water in an electric kettle for 20 minutes. Using the same mains supply he wants to reduce the boiling time of water. To do so should he increase or decrease the length of the heating element? Justify your answer.(2 Marks)	3
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Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

sqp_2021-22_term2_q12_main | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

12

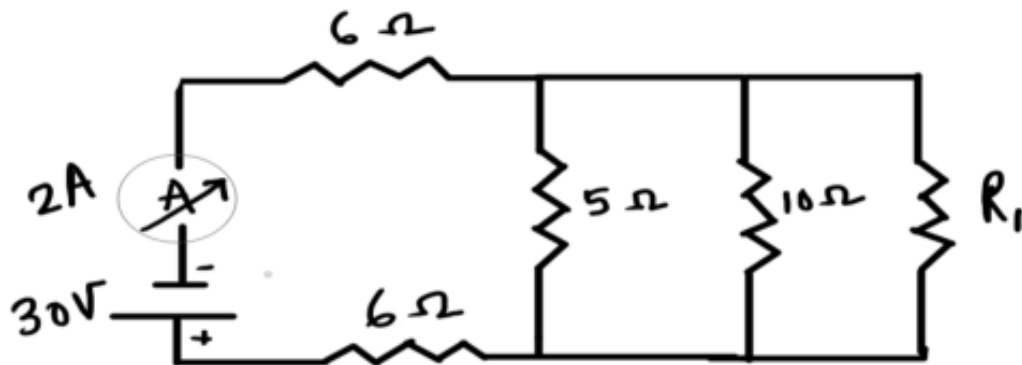


3

In the above circuit, if the current reading in the ammeter A is 2A, what would be the value of R_1 ?

Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

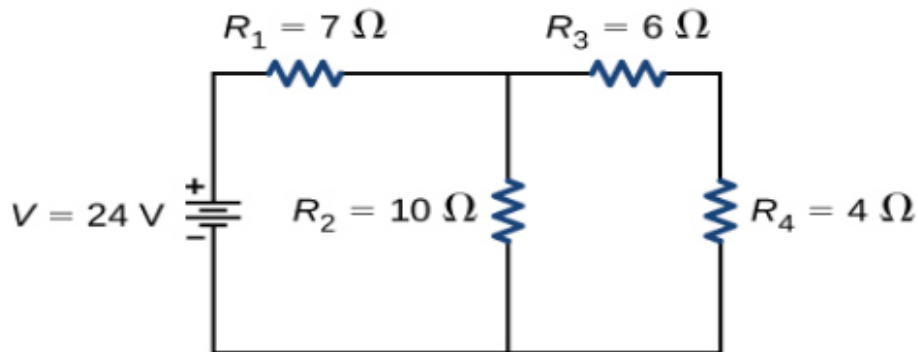
sqp_2021-22_term2_q12_main | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

sqp_2021-22_term2_q12_OR | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

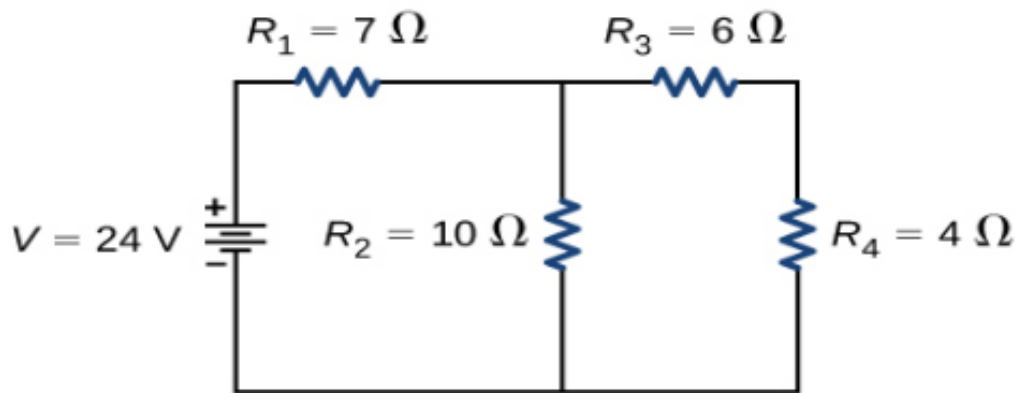
OR



Calculate the total resistance of the circuit and find the total current in the circuit.

Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

sqp_2021-22_term2_q12_OR | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



Source Q13 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 1

sqp_2022-23_q13 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

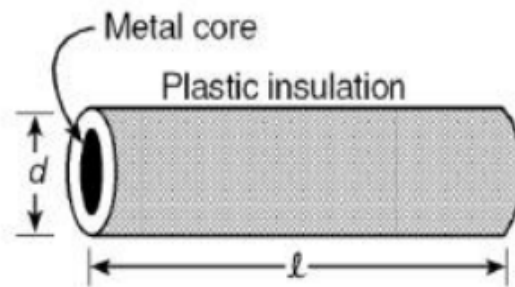
13	A complete circuit is left on for several minutes, causing the connecting copper wire to become hot. As the temperature of the wire increases, the electrical resistance of the wire (a) decreases. (b) remains the same. (c) increases. (d) increases for some time and then decreases .	1
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Source Q15 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 1

sqp_2022-23_q15 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

15

1

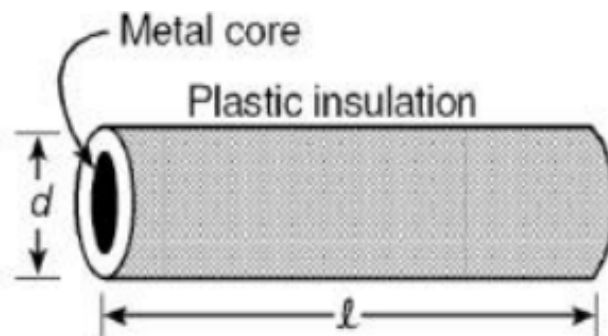


Plastic insulation surrounds a wire having diameter d and length l as shown above. A decrease in the resistance of the wire would be produced by an increase in the

- (a) length l of the wire
- (b) diameter d of the wire
- (c) temperature of the wire
- (d) thickness of the plastic insulation

Source Q15 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 1

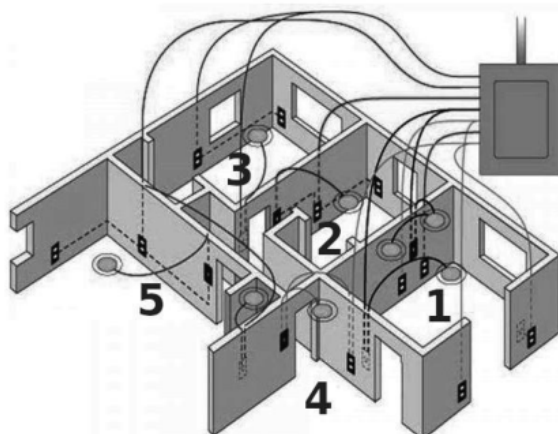
sqp_2022-23_q15 | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



Source Q36 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 5

sqp_2022-23_q36 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

36



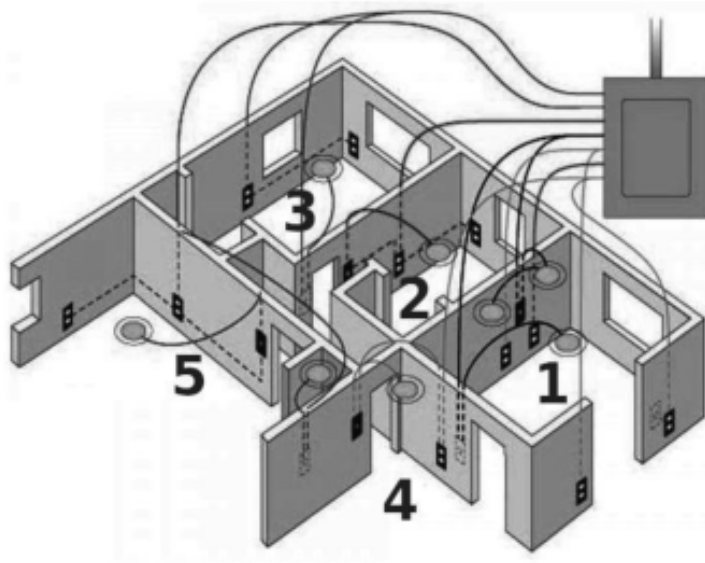
5

The diagram above is a schematic diagram of a household circuit. The house shown in the above diagram has 5 usable spaces where electrical connections are made. For this house, the mains have a voltage of 220 V and the net current coming from the mains is 22A.

- What is the mode of connection to all the spaces in the house from the mains?
- The spaces 5 and 4 have the same resistance and spaces 3 and 2 have respective resistances of 20Ω and 30Ω . Space 1 has a resistance double that of space 5. What is the net resistance for space 5.
- What is the current in space 3?
- What should be placed between the main connection and the rest of the house's electrical appliances to save them from accidental high electric current?

Source Q36 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 5

sqp_2022-23_q36 | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.



Source Q25 main | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 2

SQP_2023-24_q25_main | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

25	A piece of wire of resistance R is cut into three equal parts. These parts are then connected in parallel. If the equivalent resistance of this parallel combination is R_1 , what is the value of the ratio $R_1 : R$?	2
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Source Q32 | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 3

SQP_2023-24_q32 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

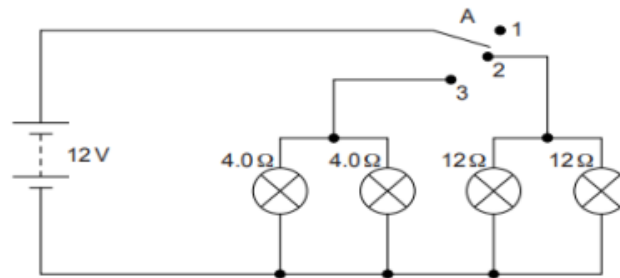
32	(i) State the law that explains the heating effect of current with respect to the measurable properties in an electrical circuit. (ii) List the factors on which the resistance of a conductor depends.	2+1
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Source Q39 iii | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4

SQP_2023-24_q39_iii | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

39

1+1+2



Vinita and Ahmed demonstrated a circuit that operates the two headlights and the two sidelights of a car, in their school exhibition. Based on their demonstrated circuit, answer the following questions.

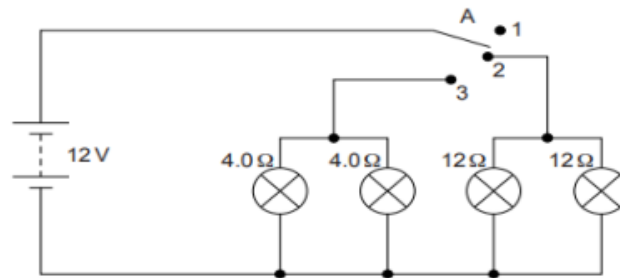
- (i) State what happens when switch A is connected to
 - a) Position 2
 - b) Position 3
- (ii) Find the potential difference across each lamp when lit.
- (iii) Calculate the current
 - a) in each 12 Ω lamp when lit.
 - b) In each 4 Ω lamp when lit.

Source Q39 OR iv | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4

SQP_2023-24_q39_OR_iv | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

39

1+1+2



Vinita and Ahmed demonstrated a circuit that operates the two headlights and the two sidelights of a car, in their school exhibition. Based on their demonstrated circuit, answer the following questions.

- (i) State what happens when switch A is connected to
 - a) Position 2
 - b) Position 3
- (ii) Find the potential difference across each lamp when lit.

Source Q39 OR iv | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4

SQP_2023-24_q39_OR_iv | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

OR

(iv) Show, with calculations, which type of lamp, $4.0\ \Omega$ or $12\ \Omega$, has the higher power.

Source Q33(ii) | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 1

SQP-2023-24-Q33-ii | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

(ii) What is the function of a fuse in a domestic circuit?

Source Q25 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

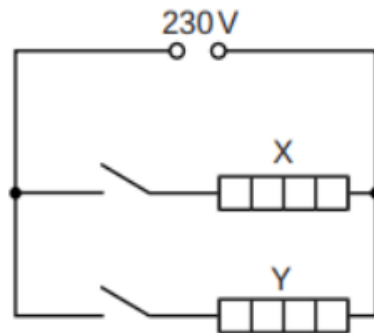
SQP_2024-25_q25_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

25

Attempt either option A or B.

2

A.



The electric circuit (above figure) in a clothes dryer contains two heaters X and Y in parallel. The above figure shows the circuit connected to a 230 V power supply. When both switches are closed, the current in X is 3.5 A.

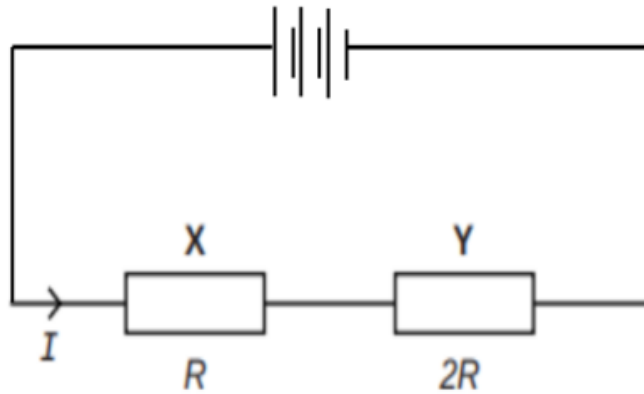
Analyse the circuit given above and answer the following questions.

- (i) Calculate the power developed in heater X.
- (ii) If the resistance of X is double that of Y calculate the current in heater Y.

Source Q25 B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

SQP_2024-25_q25_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

B.



The above figure shows two resistors X and Y connected in series to a battery. The power dissipated for this combination is P_1 . When these resistors

Source Q25 B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

SQP_2024-25_q25_B | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

are connected in parallel to the same battery then the power dissipated is given by P_2 . Find out the ratio $\frac{P_1}{P_2}$.

Source Q25 VI A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

SQP_2024-25_q25_VI_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

For visually impaired students

- A. We have four resistors A, B, C and D of resistance $3\ \Omega$, $6\ \Omega$, $9\ \Omega$ and $12\ \Omega$ respectively. Find out the lowest resistance which can be obtained by combining these four resistors.

Source Q25 VI B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

SQP_2024-25_q25_VI_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

OR

- B. You are given 2 fuse wires A and B with current ratings 2A and 5A respectively. Justify with reason which of the two would you use with a 1000W, 220V room heater?

Source Q32 | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 3

SQP_2024-25_q32 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

32	A. State the relationship between the resistance R of a wire to its length l and cross sectional area A . Use the mathematical symbols to arrive at the final formula. B. Using the formula define the resistivity of a material.	3
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Source Q36 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

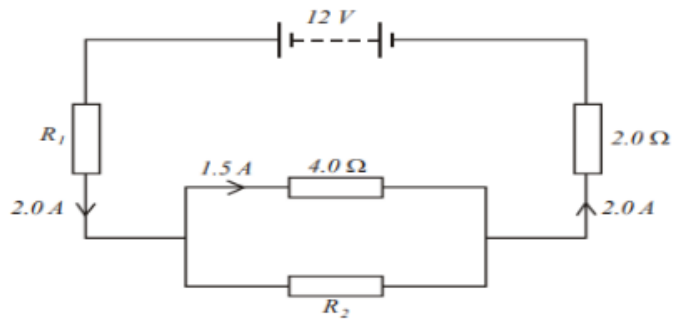
SQP_2024-25_q36_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

36

Attempt either option A or B.

5

A.



Source Q36 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

SQP_2024-25_q36_A | Crop 2 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

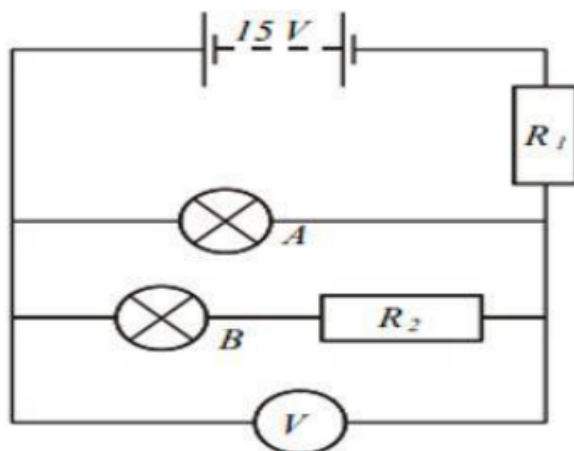
The above circuit is a part of an electrical device. Use the information given in the question to calculate the following.

- (i) Potential Difference across R_2 .
- (ii) Value of the resistance R_2 .
- (iii) Value of resistance R_1

Source Q36 B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

SQP_2024-25_q36_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

B.



As shown in the figure above A and B are two lamps. Lamp A is rated at 12 V, 24W. Lamp B is rated at 6.0 V. When lamp B operates at its rated voltage, the current in it is 3.0 A. The values of R_1 and R_2 are chosen so that both lamps operate at their rated voltages.

Based on the information given, answer the following.

- (i) Calculate the current in Lamp A.
- (ii) State and give reason for the reading of the Voltmeter.
- (iii) Calculate the resistance of R_2 .
- (iv) Find the value of the resistance R_1 .

Source Q36 VI A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

SQP_2024-25_q36_VI_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

For visually impaired students

A.

- (i) State the law and write the formula connecting the electric current flowing through a conductor and voltage applied across it.
- (ii) In a household five fans each of 100W are used for 4 hours and an electric press of 500W for 2 hours every day. Calculate the cost of using the fans and electric press for 60 days if the cost of 1 unit of electrical energy is Rs. 6.5.

Source Q36 VI B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

SQP_2024-25_q36_VI_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

OR

B.

- (i) State Joules law of heating and write its mathematical expression.
- (ii) Two resistors of resistances 2Ω and 4Ω are connected in
 - a) series
 - b) parallelwith a battery of given potential difference. Compute the ratio of total quantity of heat produced in the combination in the two cases if the total voltage and time are kept the same for both.

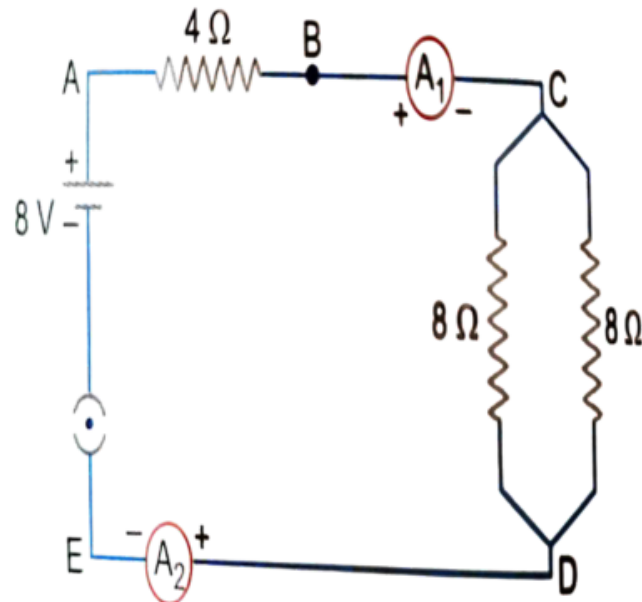
Source Q34 A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

SQP_2025-26_q34_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

34 Attempt either option A or B.

2

A.



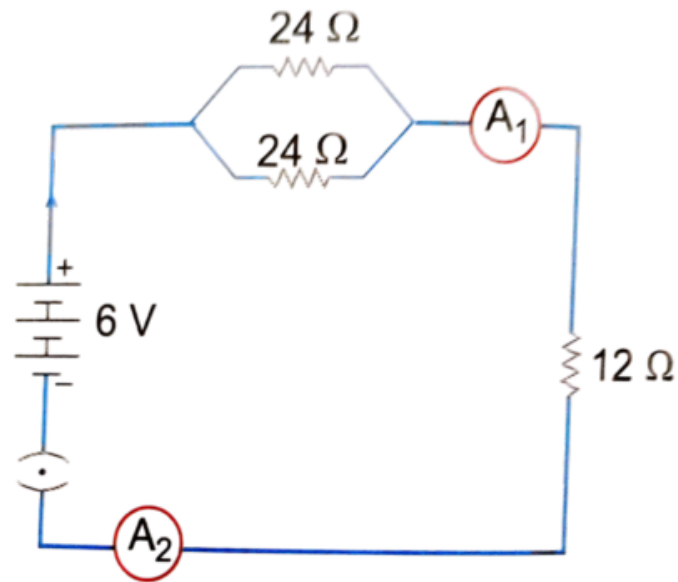
Find out the following in the electric circuit given in the figure-

- Effective resistance of two 8 ohm resistors in the combination.
- Current flowing through the 4-ohm resistor

Source Q34 B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

SQP_2025-26_q34_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

B.



Study the circuit and find out-

- (i) Current in 12 ohm resistor
- (ii) Difference in the readings of ammeter A_1 and A_2 if any

Source Q34 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

SQP_2025-26_q34_VI_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

For visually impaired students

- A. You are given four resistors each having resistance of R ohm. Find the maximum and minimum resistance that can be made with these four resistors.

Source Q34 VI B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

SQP_2025-26_q34_VI_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

OR

- B. A copper wire has a length $L=2$ m, a cross-sectional area $A=0.5$ mm², and resistivity $\rho=1.7\times 10^{-8}$ Ω -m. Calculate the resistance of another wire made of the same material whose length is twice the length of the wire but has the same cross-sectional area.

Source Q36 | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 3

SQP_2025-26_q36 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

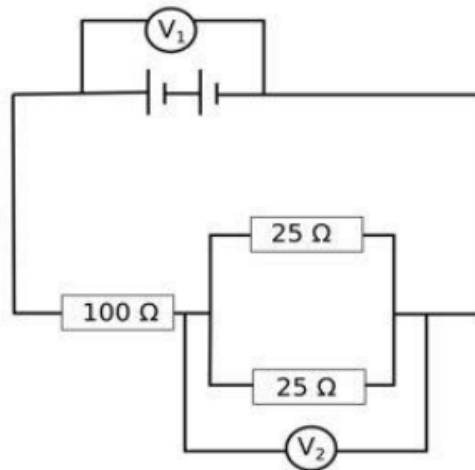
36	A student needs to make a $0.12\ \Omega$ resistor. She has some copper wire of 0.80 mm diameter. Resistivity of copper is $1.8 \times 10^{-8}\ \Omega\text{m}$ (i) Determine the cross-sectional area of the wire. (ii) Calculate the length of wire required for the $0.12\ \Omega$ resistor.	3
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Source Q39 A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

SQP_2025-26_q39_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

39 Attempt either option A or B.
A.

5



The arrangement of resistors shown in the above figure is connected to a battery.;

The power dissipation in the $100\ \Omega$ resistor is $81\ \text{W}$. Calculate

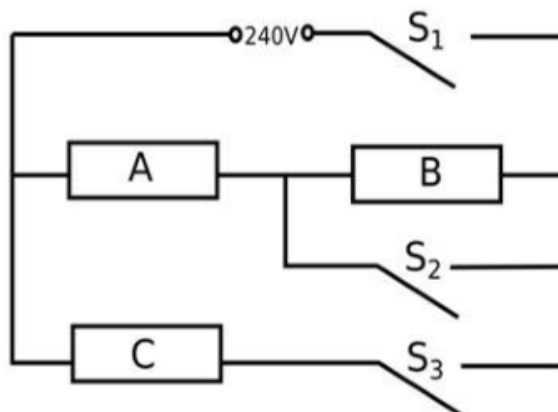
- (i) the current in the circuit
- (ii) the reading in the voltmeter V_2
- (iii) the reading in the voltmeter V_1

Source Q39 B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

SQP_2025-26_q39_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

B.

OR



An electric heater consists of three similar heating elements A, B and C, connected as shown in the figure above. Each heating element is rated as 1.2 kW, 240 V and has constant resistance. S_1 , S_2 and S_3 are respective switches.

The circuit is connected to a 240 V supply.

- (i) Calculate the resistance of one heating element.
- (ii) Calculate the current in each resistor when only S_1 and S_3 are closed.
- (iii) Calculate the power dissipated across A when S_1 , S_2 and S_3 are closed.

Source Q39 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

SQP_2025-26_q39_VI_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

For visually impaired students

A.

- (i) Explain why in household circuits only the fuse is connected in series with all the rest of the appliances but all appliances are connected in parallel to each other.
- (ii) In a household circuit, an electric heater of power 1500 W and a fan of power 500 W are connected in parallel to a 220 V supply. A fuse rated for 10 A is connected to the circuit to protect it from excessive current.
 - (a) Calculate the total current drawn by the heater and the fan.
 - (b) Determine whether the 10 A fuse is appropriate for this circuit or if it will blow.

Source Q39 VI B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

SQP_2025-26_q39_VI_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

OR

- B. Two resistors, $R_1=6\ \Omega$ and $R_2=12\ \Omega$, are connected in parallel to a 24V battery. The circuit operates for 5 minutes.
- (i) Calculate the total heat generated in both resistors.
 - (ii) If each resistor has a power rating of 100 W, determine whether it is safe to use these resistors in the circuit.

Source Q31 | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 1

SQP_2026-27_q31 | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

31	An electrical appliance rated 1000 W is operated at 250 V. What current does it draw? A. 0.25 A B. 2.5 A C. 4.0 A D. 10 A	1
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Source Q34 A | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 2

SQP_2026-27_q34_A | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

34	<u>Attempt either option A or B.</u> A. I. State Joule's Law of heating. II. What happens to the heat produced in a conductor if the resistance is doubled while keeping other variables constant?	2
----	---	---

Source Q34 B | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 2

SQP_2026-27_q34_B | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

Or

B.

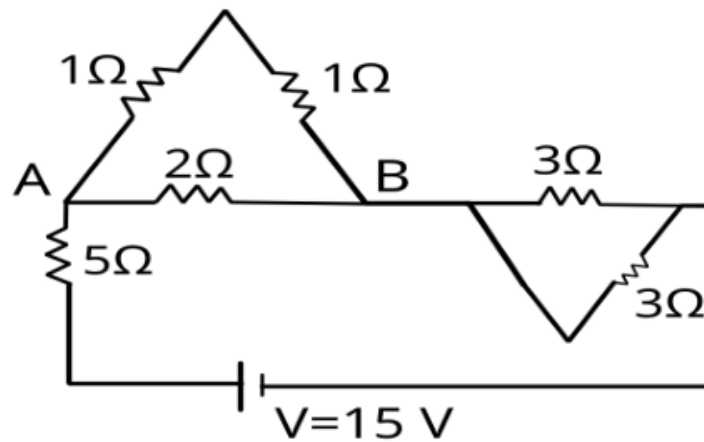
I. Define resistivity.

II. Justify with an example why resistivity is an intrinsic property of a material.

Source Q37 main | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 3

SQP_2026-27_q37_main | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

37



3

The figure shows a circuit with a 15 V battery connected to several resistors. Using the given circuit diagram:

- Find the total current supplied by the battery.
- Calculate the potential difference between points A and B.

Source Q37 VI | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 3

SQP_2026-27_q37_VI | Crop 1 | Source verification: verified | Original pixels retained; low-quality sources may need zoom.

For visually impaired students

A 15 V battery is connected in series with three resistors of 2 Ω , 3 Ω , and 5 Ω .

Calculate:

- A. The total resistance in the circuit.
- B. The total current supplied by the battery.
- C. The voltage drop across the 3 Ω resistor.

Solutions and verification status

Only explicitly verified solutions are printed. Missing solutions are not a claim that a question is unsolvable. Official marking schemes may contain errors or omissions.

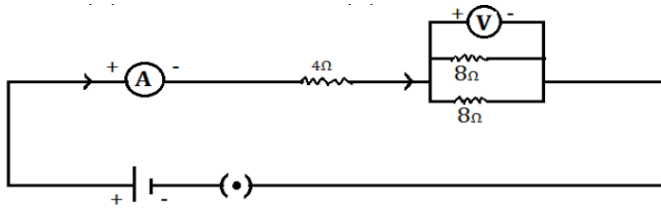
1. cbse_sqp_2017-18_electricity_q7 — Source Q7 | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 3

Circuit diagram as in MS page 1. Maximum current through $4\ \Omega$ resistor = $\sqrt{P/R} = \sqrt{16/4} = 2\text{ A}$.
Maximum current through each $8\ \Omega$ resistor = $(1/2) \times 2 = 1\text{ A}$.

Verified solution — cbse_sqp_2017-18_electricity_q7

Source Q7 | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 3 | Solution crop 1

7



1

Verified solution — cbse_sqp_2017-18_electricity_q7

Source Q7 | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 3 | Solution crop 2

$$\text{Maximum current through } 4 \, \Omega \text{ resister} = \sqrt{\frac{P}{R}} \quad 1$$

$$= \sqrt{\frac{16}{4}} = 2A$$

$$\therefore \text{Maximum current through each } 8 \, \Omega \text{ resister} = \frac{1}{2} \times 2 = 1A \quad 1$$

Marking-scheme PDF page(s), one-based: [1, 2]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

2. cbse_sqp_2017-18_electricity_q26 — Source Q26 | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 2

Official answer unavailable/withheld: Official slope solution captured, but required plotted graph absent from official MS; not a complete publishable model answer.

Answer verification note: Official slope solution captured, but required plotted graph absent from official MS; not a complete publishable model answer.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

3. cbse_sqp_2017-18_electricity_q26_OR — Source Q26 OR | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 2

1 Division = $0.5\text{A} / 10$. 17 divisions = $17/20 = 0.85\text{A}$.

Verified solution — cbse_sqp_2017-18_electricity_q26_OR

Source Q26 OR | Year: 2017-18 | Session: 2017-18 | Code: Not recorded | Marks: 2 | Solution crop 1

OR

An ammeter has 10 divisions between 0 to 0.5A. So,

$$1 \text{ Division} = 0.5\text{A} / 10$$

$$17 \text{ divisions} = 17/20 = 0.85\text{A}$$

Marking-scheme PDF page(s), one-based: [7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

**4. cbse_sqp_2018-19_electricity_q12 — Source Q12 | Year: 2018-19 |
Session: 2018-19 | Code: Not recorded | Marks: 3**

$R_1 = V^2/P_1 = 484 \, \Omega$; $R_2 = V^2/P_2 = 242 \, \Omega$. In series: $R_s = 726 \, \Omega$, $I_s = 220/726 = 10/33 \, \text{A} = 0.30 \, \text{A}$. In parallel: $R_p = 484/3 \, \Omega$, $I_p = 220 \times 3/484 = 30/22 \, \text{A} = 1.36 \, \text{A}$.

Verified solution — cbse_sqp_2018-19_electricity_q12

Source Q12 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 3 | Solution crop 1

$$12. R_1 = \frac{V^2}{P_1} = \frac{220 \times 220}{100} = 484 \, \Omega$$

3

$$R_2 = \frac{V^2}{P_2} = \frac{220 \times 220}{200} = 242 \, \Omega$$

In series:

$$R_S = R_1 + R_2 = 484 + 242 = 726 \, \Omega$$

$$\therefore I_S = \frac{V}{R_S} = \frac{220}{726} = \frac{10}{33} \, \text{A} = 0.30 \, \text{A}$$

In parallel:

$$\begin{aligned} \frac{1}{R_p} &= \frac{1}{R_1} + \frac{1}{R_2} \\ &= \frac{1}{484} + \frac{1}{242} \end{aligned}$$

$$R_p = \frac{484}{3} \, \Omega$$

$$I_P = \frac{V}{R_p} = \frac{220 \times 3}{484} = \frac{30}{22} \, \text{A} = 1.36 \, \text{A}$$

Marking-scheme PDF page(s), one-based: [4]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme:

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

**5. cbse_sqp_2018-19_electricity_q13 — Source Q13 | Year: 2018-19 |
Session: 2018-19 | Code: Not recorded | Marks: 3**

$R_a = \rho L/A$; $R_\beta = \rho(3L)/(A/3) = 9\rho L/A = 9R_a$; $R_\square = \rho(L/3)/(3A) = (1/9)\rho L/A = (1/9)R_a$. Hence $R_\beta > R_a > R_\square$. $\rho_a = \rho_\beta = \rho_\square$ because all three conductors are of the same material.

Verified solution — cbse_sqp_2018-19_electricity_q13

Source Q13 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 3 | Solution crop 1

$$13. R_a = \rho \frac{L}{A}$$

3

$$R_b = \rho \left(\frac{3L}{A/3} \right) = 9 \frac{\rho L}{A} = 9 R_a$$

$$R_c = \rho \frac{L/3}{3A} = \frac{1}{9} \frac{\rho L}{A} = \frac{1}{9} R_a$$

Hence $R_b > R_a > R_c$

$\rho_a = \rho_b = \rho_c$ because all the three conductors are of same material.

Marking-scheme PDF page(s), one-based: [4]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme:

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

**6. cbse_sqp_2018-19_electricity_q26 — Source Q26 | Year: 2018-19 |
Session: 2018-19 | Code: Not recorded | Marks: 2**

LC of ammeter = $0.2/10 \text{ A} = 0.02 \text{ A}$. LC of Voltmeter = $1/10 = 0.1 \text{ V}$. Current = $15 \times 0.02 \text{ A} = 0.3 \text{ A}$.
Potential difference = $21 \times 0.1 \text{ V} = 2.1 \text{ V}$.

Verified solution — cbse_sqp_2018-19_electricity_q26

Source Q26 | Year: 2018-19 | Session: 2018-19 | Code: Not recorded | Marks: 2 | Solution crop 1

$$26. \text{LC of ammeter} = \frac{0.2}{10} \text{ A} = 0.02 \text{ A}$$

$$\text{LC of Voltmeter} = \frac{1}{10} = 0.1 \text{ V}$$

$$\text{Current} = 15 \times 0.02 \text{ A} = 0.3 \text{ A}$$

$$\text{Potential difference} = 21 \times 0.1 \text{ V} = 2.1 \text{ V}$$

2

Marking-scheme PDF page(s), one-based: [7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme:

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

7. cbse_sqp_2019-20_electricity_q6 — Source Q6 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1

ii) $40\ \Omega$. $V = IR$; $V = 4\ \text{V}$; $I = 100\ \text{mA} = 0.1\ \text{A}$. $R = V/I = 4/0.1\ \Omega = 40\ \Omega$.

Verified solution — cbse_sqp_2019-20_electricity_q6

Source Q6 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1 | Solution crop 1

6	ii) = 40 Ω. $V=IR$, $V = 4V$, $I = 100 \text{ mA} = 0.1 \text{ A}$ Hence $R = V/I = 4/0.1 \Omega = 40 \Omega$.	1 mark
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Marking-scheme PDF page(s), one-based: [1]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

8. cbse_sqp_2019-20_electricity_q7 — Source Q7 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1

i) volt-ampere. Power = Voltage $\times$ Current.

Verified solution — cbse_sqp_2019-20_electricity_q7

Source Q7 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1 | Solution crop 1

7	i) volt-ampere Power = Voltage x Current.	1 mark
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Marking-scheme PDF page(s), one-based: [1]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

9. cbse_sqp_2019-20_electricity_q29 — Source Q29 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 5

(i) Resistivity will not change as it depends on the nature of the material of the conductor. (ii) Each part has length $L/4$; ρ and A are constant. $R = \rho L/A$; resistance of each part = $R/4$. (a) In parallel, equivalent resistance = $R/16 \Omega$. (b) In series, equivalent resistance = $R \Omega$. (iii) $P = V^2/R$. At a given voltage, the parallel combination has smaller equivalent resistance and therefore consumes more power.

Verified solution — cbse_sqp_2019-20_electricity_q29

Source Q29 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 5 | Solution crop 1

29	(i) Resistivity will not change as it depends on the nature of the material of the conductor. (ii) The length of each part becomes $L/4$. ρ, A constant. $R = \rho L / A$. Resistance of each part = $R_{\text{part}} = (\rho L / 4) / A = R / 4$.	1 mark 0.5 mark
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Verified solution — cbse_sqp_2019-20_electricity_q29

Source Q29 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 5 | Solution crop 2

(a) In parallel the $\frac{1}{R_{eq}} = \frac{1}{R/4} + \frac{1}{R/4} + \frac{1}{R/4} + \frac{1}{R/4} = \frac{4}{R/4} = \frac{16}{R} \rightarrow R_{eq} = R/16 \Omega$

0.5 mark

(b) In series the $R_{eq} = R/4 + R/4 + R/4 + R/4 = R \Omega$

1 mark

(iii) $P = V^2/R$.

1 mark

If R_{eq} is less, power consumed will be more.

0.5 mark

In the given case, R_{eq} is lesser in the parallel and power consumed will be more.

0.5 mark

Marking-scheme PDF page(s), one-based: [6, 7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

**10. SQP-2019-20-Q14 — Source Q14 | Year: 2019-20 | Session: 2019-20 |
Code: Not recorded | Marks: 1**

iv) Assertion is false but reason is true.

Verified solution — SQP-2019-20-Q14

Source Q14 | Year: 2019-20 | Session: 2019-20 | Code: Not recorded | Marks: 1 | Solution crop 1

14	iv) Assertion is false but reason is true.	1 mark
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Marking-scheme PDF page(s), one-based: [2]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

**11. sqp_2020-21_q9_main — Source Q9 | Year: 2020-21 | Session: 2020-21
| Code: Not recorded | Marks: 1**

Voltage-drop is same across both.

Verified solution — sqp_2020-21_q9_main

Source Q9 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 1 | Solution crop 1

9	Voltage-drop is same across both OR $W=QV$	1
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Marking-scheme PDF page(s), one-based: [1]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

12. sqp_2020-21_q9_OR — Source Q9 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 1

$W = QV$.

Verified solution — sqp_2020-21_q9_OR

Source Q9 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 1 | Solution crop 1

9	Voltage-drop is same across both OR $W=QV$	1
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Marking-scheme PDF page(s), one-based: [1]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

13. sqp_2020-21_q26 — Source Q26 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 2

The overall current needed = 9 A and the voltage is 12 V. By Ohm's law, $R = 12/9 = 4/3 \Omega$. R1 and R2 are in parallel, so $R = R_1 R_2 / (R_1 + R_2) = 4 R_2 / (4 + R_2) = 4/3$. Hence $R_2 = 2 \Omega$.

Verified solution — sqp_2020-21_q26

Source Q26 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 2 | Solution crop 1

26	The overall current needed = 9A. The voltage is 12V Hence by Ohm's Law $V=IR$, The resistance for the entire circuit = $12/9 = 4/3 \Omega$. = R R_1 and R_2 are in parallel. Hence, $R=(R_1 R_2)/(R_1 + R_2) = 4R_2/(4+R_2) = 4/3$ $R_2 = 2\Omega$	2
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Marking-scheme PDF page(s), one-based: [4]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

14. sqp_2020-21_q36_main — Source Q36 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 5

(i) The lamps are in parallel. (ii) A faulty lamp does not affect the working of the others; all lamps use the full potential of the battery. (iii) The lamp with the highest power glows brightest. All have the same voltage; C has the largest current. $PC = 5 \times 60 = 300 \text{ W}$, the maximum. (iv) Total current $= 3 + 4 + 5 + 3 = 15 \text{ A}$; $R = V/I = 60/15 = 4 \Omega$. The MS prints the final unit incorrectly as A.

Source Q36 | Year: 2020-21 | Session: 2020-21 | Code: Not recorded | Marks: 5 | Solution crop 1

Electricity | Official-source archive

Marking-scheme PDF page(s), one-based: [8]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

15. sqp_2021-22_term2_q11 — Source Q11 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

(a) $E = P \times T = 3 \times 2 = 6$ kWh. Cost = $6 \times 3.50 = \text{Rs. } 21.0$. (b) To reduce the boiling time, the rate of heat production should be large. $P = V^2/R$; V is constant, so R should be decreased. Since R is directly proportional to length l , the length should be decreased.

Verified solution — sqp_2021-22_term2_q11

Source Q11 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3 | Solution crop 1

11	a. $E = P \times T$ SO, $E = 3 \times 2 = 6 \text{ kWh}$ Cost of buying electricity from the main electricity board = $6 \times 3.50 =$ Rs. 21.0	(1 mark)	3
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Verified solution — sqp_2021-22_term2_q11

Source Q11 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3 | Solution crop 2

-
- b. To reduce the boiling time using the same mains supply, the rate of heat production should be large. We know that $P = V^2/R$. Since V is constant, R should be decreased. Since R is directly proportional to l so length should be decreased. (2 marks)

Marking-scheme PDF page(s), one-based: [3, 4]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-MS_Term2.pdf

16. sqp_2021-22_term2_q12_main — Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

$R_1 = 30\ \Omega$. The parallel block has $1/R_p = 1/5 + 1/10 + 1/R_1$, so $R_p = 10R_1/(3R_1 + 10)$. The two $6\ \Omega$ resistors are in series with that block. $R_{eq} = 30/2 = 15\ \Omega$, hence $12 + 10R_1/(3R_1 + 10) = 15$ and $R_1 = 30\ \Omega$.

Verified solution — sqp_2021-22_term2_q12_main

Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3 | Solution crop 1

12

5 ohm, 10 ohm and R_1 are in series

$$1/R_p = 1/5 + 1/10 + 1/R_1$$

$$1/R_p = (2+1)/10 + 1/R_1$$

$$= 3/10 + 1/R_1$$

$$1/R_p = (3 R_1 + 10)/10 R_1$$

$$R_p = 10 R_1 / (3 R_1 + 10)$$

Now, 6 ohm, 6 ohm and R_p are in series

Thus,

$$R_{eq} = 12 + 10 R_1 / (3 R_1 + 10) \text{ ----- (1)} \quad (1 \text{ mark})$$

$$V = I R_{eq}$$

From the circuit

$$R_{eq} = 30/2 = 15 \text{ A} \text{ ----- (2)} \quad (1 \text{ mark})$$

Equating (1) and (2)

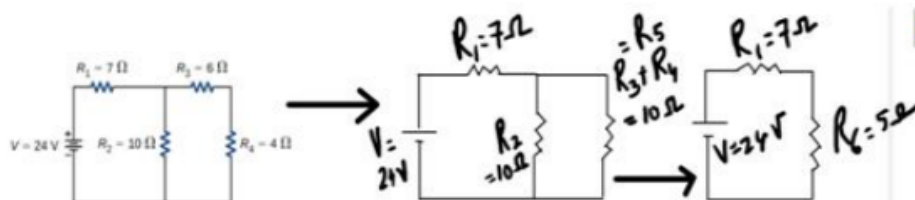
$$12 + 10 R_1 / (3 R_1 + 10) = 15$$

$$10 R_1 / (3 R_1 + 10) = 3$$

$$10 R_1 = (9 R_1 + 30)$$

$$\text{Thus, } R_1 = 30 \text{ ohm.} \quad (1 \text{ mark})$$

OR

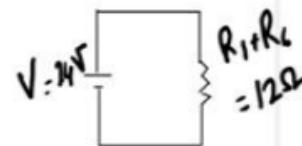


R_3 and R_4 are in series, hence the equivalent resistance of those two = $R_5 = R_3 + R_4 = 10 \text{ ohms}$. **0.5 marks**

R_5 and R_2 are in parallel. Let R_6 be the equivalent resistance for them. Hence $R_6 = (R_5 \cdot R_2) / (R_5 + R_2) = 100/20 = 5 \text{ ohms}$. **0.5 marks**

Now R_1 and R_6 are in series and hence the final equivalent resistance of the entire circuit is $R = R_1 + R_6 = 12 \text{ ohms}$. **1 mark**

By Ohm's Law we know that $V = IR$, hence $I = V/R$. Hence the current in the circuit is $24/12 \text{ A} = 2 \text{ A}$ (Final Answer) **1 mark**



(0.5+0.5+1+1)

Marking-scheme PDF page(s), one-based: [4]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-MS_Term2.pdf

17. sqp_2021-22_term2_q12_OR — Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3

R3 and R4 are in series: $R_5 = 6 + 4 = 10 \Omega$. R5 and R2 are in parallel: $R_6 = (10 \times 10)/(10 + 10) = 5 \Omega$. R1 and R6 are in series: total resistance = $7 + 5 = 12 \Omega$. With $V = 24 \text{ V}$, $I = V/R = 24/12 = 2 \text{ A}$.

Verified solution — sqp_2021-22_term2_q12_OR

Source Q12 | Year: 2021-22 | Session: 2021-22 | Code: Not recorded | Marks: 3 | Solution crop 1

12

5 ohm, 10 ohm and R_1 are in series

$$1/R_p = 1/5 + 1/10 + 1/R_1$$

$$1/R_p = (2+1)/10 + 1/R_1$$

$$= 3/10 + 1/R_1$$

$$1/R_p = (3 R_1 + 10)/10 R_1$$

$$R_p = 10 R_1 / (3 R_1 + 10)$$

Now, 6 ohm, 6 ohm and R_p are in series

Thus,

$$R_{eq} = 12 + 10 R_1 / (3 R_1 + 10) \text{ ----- (1)} \quad (1 \text{ mark})$$

$$V = I R_{eq}$$

From the circuit

$$R_{eq} = 30/2 = 15 \text{ A} \text{ ----- (2)} \quad (1 \text{ mark})$$

Equating (1) and (2)

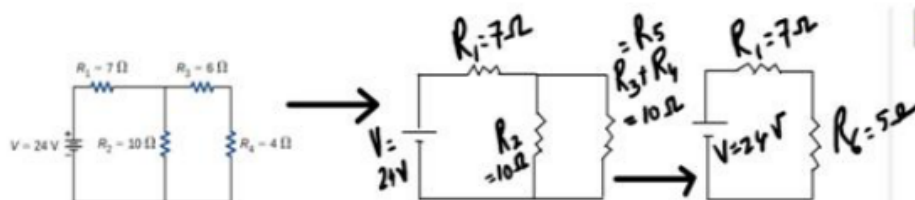
$$12 + 10 R_1 / (3 R_1 + 10) = 15$$

$$10 R_1 / (3 R_1 + 10) = 3$$

$$10 R_1 = (9 R_1 + 30)$$

$$\text{Thus, } R_1 = 30 \text{ ohm.} \quad (1 \text{ mark})$$

OR

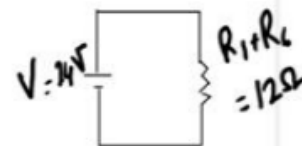


R_3 and R_4 are in series, hence the equivalent resistance of those two = $R_5 = R_3 + R_4 = 10 \text{ ohms}$. **0.5 marks**

R_5 and R_2 are in parallel. Let R_6 be the equivalent resistance for them. Hence $R_6 = (R_5 \cdot R_2) / (R_5 + R_2) = 100/20 = 5 \text{ ohms}$ **0.5 marks**

Now R_1 and R_6 are in series and hence the final equivalent resistance of the entire circuit is $R = R_1 + R_6 = 12 \text{ ohms}$. **1 mark**

By Ohm's Law we know that $V = IR$, hence $I = V/R$. Hence the current in the circuit is $24/12 \text{ A} = 2 \text{ A}$ (Final Answer) **1 mark**



(0.5+0.5+1+1)

Marking-scheme PDF page(s), one-based: [4]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-MS_Term2.pdf

**18. sqp_2022-23_q13 — Source Q13 | Year: 2022-23 | Session: 2022-23 |
Code: Not recorded | Marks: 1**

(c) increases

Verified solution — sqp_2022-23_q13

Source Q13 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 1 | Solution crop 1

13.	(c) increases	1
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Marking-scheme PDF page(s), one-based: [1]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

**19. sqp_2022-23_q15 — Source Q15 | Year: 2022-23 | Session: 2022-23 |
Code: Not recorded | Marks: 1**

(b) diameter d of the wire

Verified solution — sqp_2022-23_q15

Source Q15 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 1 | Solution crop 1

15.	(b) diameter d of the wire	1
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Marking-scheme PDF page(s), one-based: [1]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

20. sqp_2022-23_q36 — Source Q36 | Year: 2022-23 | Session: 2022-23 | Code: Not recorded | Marks: 5

Official answer unavailable/withheld: Official MS omits c,d; independent c derivation labelled; complete answer not verified.

Answer verification note: Official MS omits c,d; independent c derivation labelled; complete answer not verified.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

21. SQP_2023-24_q25_main — Source Q25 main | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 2

Resistance of each part is $R/3 \Omega$ (as resistance is proportional to the length of the wire.) [0.5]

$$1/R_1 = 3/R + 3/R + 3/R = 9/R \quad [1]$$

$$\therefore R_1 = R/9 \therefore R_1/R = 1/9 \quad [0.5]$$

Verified solution — SQP_2023-24_q25_main

Source Q25 main | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 2 | Solution crop 1

25	Resistance of each part is $\frac{R}{3} \Omega$ (as resistance is proportional to the length of the wire.) $\frac{1}{R_1} = \frac{3}{R} + \frac{3}{R} + \frac{3}{R} = \frac{9}{R}$ $\therefore R_1 = \frac{R}{9} \therefore \frac{R_1}{R} = \frac{1}{9}$ OR	2
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Marking-scheme PDF page(s), one-based: [2]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

22. SQP_2023-24_q32 — Source Q32 | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 3

Official answer unavailable/withheld: Official answer retains erroneous digit 1 in $H=1^2Rt$; correct formula explained in notes.

Answer verification note: Official answer retains erroneous digit 1 in $H=1^2Rt$; correct formula explained in notes.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

23. SQP_2023-24_q39_iii — Source Q39 iii | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4

(i) 12 Ω lamps (only) on.

(a) 4 Ω lamps (only) on [0.5+0.5]

(ii) 12 V for both sets of lamps and all of them are in parallel. [1]

(iii) 12 Ω lamps are on when the wire is connected to position 2. Voltage across both 12 Ω lamps = 12 V. $V=IR$ (Ohm's law). $I = V/R = 12/12 = 1\text{A}$. [1]

4 Ω lamps are on when the wire is connected to position 3. Voltage across both 4 Ω lamps = 12 V. $V=IR$ (Ohm's law). $I = V/R = 12/4 = 3\text{A}$. [1]

Verified solution — SQP_2023-24_q39_iii

Source Q39 iii | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4 | Solution crop 1

39	(i) 12 Ω lamps (only) on. (a) 4 Ω lamps (only) on [0.5+0.5] (ii) 12 V for both sets of lamps and all of them are in parallel. [1] (iii) 12 Ω lamps are on when the wire is connected to position 2. Voltage across both 12 Ω lamps = 12 V. $V=IR$ (Ohm's law). $I = \frac{V}{R} = \frac{12}{12} = 1\text{A}.$ [1] 4 Ω lamps are on when the wire is connected to position 3. Voltage across both 4 Ω lamps = 12 V. $V=IR$ (Ohm's law). $I = \frac{V}{R} = \frac{12}{4} = 3\text{A}.$ [1]	4
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Marking-scheme PDF page(s), one-based: [6]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

24. SQP_2023-24_q39_OR_iv — Source Q39 OR iv | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4

(i) 12 Ω lamps (only) on.

(a) 4 Ω lamps (only) on [0.5+0.5]

(ii) 12 V for both sets of lamps and all of them are in parallel. [1]

OR

$$P = V^2/R$$

All lamps are in parallel and hence same V for all lamps.

For 4 Ω lamps $\rightarrow P = (12 \times 12)/4 = 36 \text{ W}$

For 12 Ω lamps $\rightarrow P = (12 \times 12)/12 = 12 \text{ W}$

Hence 4 Ω lamps will have higher power. [0.5 x 4]

Verified solution — SQP_2023-24_q39_OR_iv

Source Q39 OR iv | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4 | Solution crop 1

39	(i) 12 Ω lamps (only) on. (a) 4 Ω lamps (only) on (ii) 12 V for both sets of lamps and all of them are in parallel.	[0.5+0.5] [1]	4
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Verified solution — SQP_2023-24_q39_OR_iv

Source Q39 OR iv | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 4 | Solution crop 2

OR

$$P = V^2 / R$$

All lamps are in parallel and hence same V for all lamps.

$$\text{For } 4 \, \Omega \text{ lamps} \rightarrow P = \frac{12 \times 12}{4} = 36 \, W$$

$$\text{For } 12 \, \Omega \text{ lamps} \rightarrow P = \frac{12 \times 12}{12} = 12 \, W$$

Hence 4 Ω lamps will have higher power.

[0.5 x 4]

Marking-scheme PDF page(s), one-based: [6]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

25. SQP-2023-24-Q33-ii — Source Q33(ii) | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 1

An electrical fuse is a safety device that operates to provide protection against the overflow of current in an electrical circuit. An important component of an electrical fuse is a metal wire or strip that melts when excess current flows through it.

Verified solution — SQP-2023-24-Q33-ii

Source Q33(ii) | Year: 2023-24 | Session: 2023-24 | Code: Not recorded | Marks: 1 | Solution crop 1

- (ii) An electrical fuse is a safety device that operates to provide protection against the overflow of current in an electrical circuit. An important component of an electrical fuse is a metal wire or strip that melts when excess current flows through it. [1]

Marking-scheme PDF page(s), one-based: [3]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

26. SQP_2024-25_q25_A — Source Q25 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

A. $P = VI$ (1)

$$= 230 \times 3.5 = 805 \text{ W}$$

$I \propto 1/R$, so half the resistance means double the current.

Therefore, current in Y = 7.0 A. (1)

Verified solution — SQP_2024-25_q25_A

Source Q25 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2 | Solution crop 1

25	<u>Student to attempt either A or B.</u>	2
	A. $P = VI$	(1)
	$= 230 \times 3.5 = 805 \text{ W}$	
	$I \propto \frac{1}{R}$, so half the resistance means double the current.	
	Therefore, current in Y = 7.0 A.	(1)

Marking-scheme PDF page(s), one-based: [3]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

27. SQP_2024-25_q25_B — Source Q25 B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

B. For series total resistance is $R+2R = 3R$ (0.5)

$$P_1 = V^2/(3R). \text{ (0.5)}$$

For parallel total Resistance is $2R/3$. (0.5)

$$P_2 = V^2/(2R/3) = 3V^2/(2R).$$

$$P_1/P_2 = 2/9. \text{ (0.5)}$$

Verified solution — SQP_2024-25_q25_B

Source Q25 B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2 | Solution crop 1

B. For series total resistance is $R+2R = 3R$

(0.5)

$$P_1 = \frac{V^2}{3R}.$$

(0.5)

For parallel total Resistance is $\frac{2R}{3}$.

(0.5)

$$P_2 = \frac{V^2}{2R/3} = \frac{3V^2}{2R}.$$

$$\frac{P_1}{P_2} = \frac{2}{9}.$$

(0.5)

Marking-scheme PDF page(s), one-based: [3]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

28. SQP_2024-25_q25_VI_A — Source Q25 VI A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

Official answer unavailable/withheld: Official arithmetic answer 1.5 ohm is erroneous; correct 1.44 ohm documented separately.

Answer verification note: Official arithmetic answer 1.5 ohm is erroneous; correct 1.44 ohm documented separately.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

29. SQP_2024-25_q25_VI_B — Source Q25 VI B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2

B. $P = V \times I$ (0.5)

$I = P/V = 1000/220 = 4.54 \text{ A}$ (0.5)

We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)

Verified solution — SQP_2024-25_q25_VI_B

Source Q25 VI B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 2 | Solution crop 1

OR

$$\text{B. } P = V \times I \quad (0.5)$$

$$I = \frac{P}{V} = \frac{1000}{220} = 4.54 \text{ A} \quad (0.5)$$

We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)

Marking-scheme PDF page(s), one-based: [3]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

30. SQP_2024-25_q32 — Source Q32 | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 3

A. $R \propto l$

$R \propto l/A$

$R = \rho l/A$. (1.5)

B. $R = \rho l/A$

$\rho = R A/l$ for $A = 1\text{m}^2$ and $l = 1\text{m}$, we have

$\rho = R$

Hence resistivity is the resistance offered by a wire of length 1 m having a cross sectional area of 1 m^2 . (1.5)

Verified solution — SQP_2024-25_q32

Source Q32 | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 3 | Solution crop 1

32	A. $R \propto l$ $R \propto \frac{l}{A}$ $R = \rho \frac{l}{A}. \quad (1.5)$ B. $R = \rho \frac{l}{A}$ $\rho = R \frac{A}{l} \text{ for } A = 1\text{m}^2 \text{ and } l = 1\text{m, we have}$ $\rho = R$ Hence resistivity is the resistance offered by a wire of length 1 m having a cross sectional area of 1 m². (1.5)	3
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Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

31. SQP_2024-25_q36_A — Source Q36 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

A.

(i) p.d. across $4\ \Omega$ resistor = p.d. Across R_2 as both are in parallel. (0.5)

$$1.5\text{ (A)} \times 4\text{ (}\Omega\text{)} = 6\text{ V (0.5)}$$

(ii) Total Current through $4\ \Omega$ and $R_2 = 2.0\text{ A}$ (given).

Current through $4\ \Omega = 1.5\text{ A}$ (given) (0.5)

Hence current through $R_2 = 2 - 1.5 = 0.5\text{ A}$

Using Ohm's law for R_2 we get (0.5)

$$6\text{ V} = 0.5\text{ A} \times R_2$$

$$\text{Hence } R_2 = 6/0.5 = 12\ \Omega\text{ (0.5)}$$

(iii) p.d. across $R_1 = \text{Total p.d.} - (\text{p.d. across } R_2) - (\text{p.d. across } 2.0\ \Omega)\text{ (0.5)}$

$$\text{p.d. across } 2.0\ \Omega = 2 \times 2 = 4\text{ V}$$

p.d. across $R_2 = 6\text{ V}$ (calculated before) (0.5)

$$\text{Hence p.d. across } R_1 = 12 - 6 - 4 = 2\text{ V (0.5)}$$

Current through $R_1 = 2\text{ A}$ (0.5)

Using Ohm's Law, we get

$$R_1 = 2\text{ V}/2\text{ A} = 1\ \Omega\text{ (0.5)}$$

Alternative method

$$\text{Total Resistance} = R_1 + (4 \times 12)/(4+12) + 2 = 12\text{ V}/2\text{ A} = 6\ \Omega\text{ (0.5)}$$

$$R_1 = 6 - (3 + 2) = 1\ \Omega\text{ (0.5)}$$

Verified solution — SQP_2024-25_q36_A

Source Q36 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5 | Solution crop 1

36	<u>Students to attempt either option A or B.</u>	5
	A.	
	(i) p.d. across $4\ \Omega$ resistor = p.d. Across R_2 as both are in parallel.	(0.5)
	$1.5(A) \times 4\ (\Omega) = 6\ V$	(0.5)
	(ii) Total Current through $4\ \Omega$ and $R_2 = 2.0\ A$ (given).	
	Current through $4\ \Omega = 1.5\ A$ (given)	(0.5)
	Hence current through $R_2 = 2 - 1.5 = 0.5\ A$	

Verified solution — SQP_2024-25_q36_A

Source Q36 A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5 | Solution crop 2

- | | | |
|-------|---|-------|
| | Using Ohm's law for R_2 we get | (0.5) |
| | $6\text{ V} = 0.5\text{ A} \times R_2$ | |
| | Hence $R_2 = 6/0.5 = 12\ \Omega$ | (0.5) |
| (iii) | p.d. across $R_1 = \text{Total p.d.} - (\text{p.d. across } R_2) - (\text{p.d. across } 2.0\ \Omega)$ | |
| | (0.5) | |
| | p.d. across $2.0\ \Omega = 2 \times 2 = 4\text{ V}$ | |
| | p.d. across $R_2 = 6\text{ V}$ (calculated before) | (0.5) |
| | Hence p.d. across $R_1 = 12 - 6 - 4 = 2\text{ V}$ | (0.5) |
| | Current through $R_1 = 2\text{ A}$ | (0.5) |
| | Using Ohm's Law, we get | |
| | $R_1 = 2\text{ V}/2\text{ A} = 1\ \Omega$ | (0.5) |

Alternative method

Total Resistance = $R_1 + \frac{(4 \times 12)}{(4+12)} + 2 = 12\text{ V}/2\text{ A} = 6\ \Omega$	(0.5)
$R_1 = 6 - (3 + 2) = 1\ \Omega$	(0.5)

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

32. SQP_2024-25_q36_B — Source Q36 B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

B.

(i) Use of $P = IV$ (0.5)

$$I = P \div V = 24 \text{ W} \div 12 \text{ V}$$

Current in lamp A = 2 A (0.5)

(ii) Voltmeter reading = 12 V (0.5)

Lamp A and Lamp B are in parallel.

Hence p.d. across the arm containing A = p.d. across arm containing B
= 12 V (from a) (0.5)

(iii) p.d. across R_2 + p.d. across B = 12 V. (0.5)

p.d. across B = 6 V (given)

Hence p.d. across R_2 = 12 V - 6 V = 6V (0.5)

Current through R_2 = Current through B = 3A (given) (0.5)

Use of $R = V/I$

$$R_2 = 6\text{V}/3\text{A} = 2\Omega \text{ (0.5)}$$

(iv) Current through R_1 = Total Current = 3A+2A = 5A

p.d. across R_1 = 15V - 12 V = 3V (0.5)

$$R_1 = 3\text{V}/5\text{A} = 0.6 \Omega \text{ (0.5)}$$

Verified solution — SQP_2024-25_q36_B

Source Q36 B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5 | Solution crop 1

B.		
(i)	Use of $P = IV$ $I = P \div V = 24 \text{ W} \div 12\text{V}$ Current in lamp A – 2 A	(0.5) (0.5)
(ii)	Voltmeter reading = 12 V Lamp A and Lamp B are in parallel. Hence p.d. across the arm containing A = p.d. across arm containing B = 12 V (from a)	(0.5) (0.5)
(iii)	p.d. across R_2 + p.d. across B = 12 V. p.d. across B = 6 V (given) Hence p.d. across $R_2 = 12 \text{ V} - 6 \text{ V} = 6\text{V}$	(0.5) (0.5)
	Current through $R_2 =$ Current through B = 3A (given) Use of $R = V/I$ $R_2 = 6\text{V}/3\text{A} = 2\Omega$	(0.5) (0.5)
(iv)	Current through $R_1 =$ Total Current = $3\text{A} + 2\text{A} = 5\text{A}$ p.d. across $R_1 = 15\text{V} - 12 \text{ V} = 3\text{V}$ $R_1 = 3\text{V}/5\text{A} = 0.6 \Omega$	(0.5) (0.5)

Marking-scheme PDF page(s), one-based: [9]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

33. SQP_2024-25_q36_VI_A — Source Q36 VI A | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

Official answer unavailable/withheld: Official energy cost omits four fans; correct Rs1170 documented separately.

Answer verification note: Official energy cost omits four fans; correct Rs1170 documented separately.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

34. SQP_2024-25_q36_VI_B — Source Q36 VI B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5

- B.
- (i) Joule's Law of Heating states that the amount of heat produced in a conductor is directly proportional to the square of the electric current passing through it, the resistance of the conductor, and the time for which the current flows. Mathematically, it can be expressed as
- $$H = I^2 R t$$
- H is the heat produced (in joules),
I is the electric current (in amperes),
R is the resistance of the conductor (in ohms),
t is the time for which the current flows (in seconds).
- (ii) Let the equivalent resistance in series be denoted by R_s and that for parallel be denoted by R_p . Total voltage of the circuit is given by V in both cases and the time is denoted by t.
- $$R_s = 2 + 4 = 6\Omega$$
- $$1/R_p = 1/2 + 1/4 = 3/4, R_p = 4/3$$
- $$H_s = (V^2/R_s)t, H_p = (V^2/R_p)t$$
- $$H_s/H_p = R_p/R_s = 2/9$$

Verified solution — SQP_2024-25_q36_VI_B

Source Q36 VI B | Year: 2024-25 | Session: 2024-25 | Code: Not recorded | Marks: 5 | Solution crop 1

OR

B.

- (i) Joule's Law of Heating states that the amount of heat produced in a conductor is directly proportional to the square of the electric current passing through it, the resistance of the conductor, and the time for which the current flows. Mathematically, it can be expressed as

$$H = I^2 R t$$

- H is the heat produced (in joules),
 - I is the electric current (in amperes),
 - R is the resistance of the conductor (in ohms),
 - t is the time for which the current flows (in seconds).
- (ii) Let the equivalent resistance in series be denoted by R_s and that for parallel be denoted by R_p . Total voltage of the circuit is given by V in both cases and the time is denoted by t .
- $R_s = 2 + 4 = 6\Omega$
 - $\frac{1}{R_p} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, R_p = \frac{4}{3}$
 - $H_s = \frac{V^2}{R_s} t, H_p = \frac{V^2}{R_p} t$
 - $\frac{H_s}{H_p} = \frac{R_p}{R_s} = \frac{2}{9}$

Marking-scheme PDF page(s), one-based: [10]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

35. SQP_2025-26_q34_A — Source Q34 A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

A.

(i) $R = R_1 R_2 / (R_1 + R_2) = (8 \times 8) / (8 + 8) = 4 \text{ ohms}$

(ii) $I = V/R = 8/(4+4) = 1 \text{ A}$

Verified solution — SQP_2025-26_q34_A

Source Q34 A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2 | Solution crop 1

34	<u>Student to attempt either A or B.</u> A. (i) $R = \frac{R_1 R_2}{R_1 + R_2} = \frac{8 \times 8}{8 + 8} = 4 \text{ ohms}$ (ii) $I = \frac{V}{R} = \frac{8}{(4 + 4)} = 1 \text{ A}$ OR	2
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Marking-scheme PDF page(s), one-based: [7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

36. SQP_2025-26_q34_B — Source Q34 B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

B.

(i) $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{24} + \frac{1}{24} = \frac{2}{24}$

$R_P = 12 \text{ ohms}$

$R_T = R_P + 12 = 24 \text{ ohms}$

$I = V/R = 6/24 = 0.25 \text{ A}$

(ii) Same readings of A_1 and A_2

Verified solution — SQP_2025-26_q34_B

Source Q34 B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2 | Solution crop 1

B.

(i)

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{24} + \frac{1}{24} = \frac{2}{24}$$

$$R_p = 12 \text{ ohms}$$

$$R_T = R_p + 12 = 24 \text{ ohms}$$

$$I = \frac{V}{R} = \frac{6}{24} = 0.25 \text{ A}$$

(ii) Same readings of A_1 and A_2

Marking-scheme PDF page(s), one-based: [7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

37. SQP_2025-26_q34_VI_A — Source Q34 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

(i) Maximum Resistance:

To get the maximum resistance, connect all four resistors in series.

The total resistance R_{max} in series is the sum of the individual resistance:

$$R_{\text{max}} = R + R + R + R = 4R$$

(ii) Minimum Resistance:

To get the minimum resistance, connect all four resistors in parallel.

The total resistance R_{min} in parallel is given by:

$$1/R_{\text{min}} = 1/R + 1/R + 1/R + 1/R = 4/R \Rightarrow R_{\text{min}} = R/4$$

Verified solution — SQP_2025-26_q34_VI_A

Source Q34 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2 | Solution crop 1

For visually impaired students

A.

(i) Maximum Resistance:

- To get the maximum resistance, connect all four resistors in series.
- The total resistance R_{max} in series is the sum of the individual resistance:

$$R_{max} = R + R + R + R = 4R$$

(ii) Minimum Resistance:

- To get the minimum resistance, connect all four resistors in parallel.

Verified solution — SQP_2025-26_q34_VI_A

Source Q34 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2 | Solution crop 2

- The total resistance R_{min} in parallel is given by:

$$\frac{1}{R_{min}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{4}{R}$$

$$R_{min} = \frac{R}{4}$$

Marking-scheme PDF page(s), one-based: [7, 8]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

38. SQP_2025-26_q34_VI_B — Source Q34 VI B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2

Given:

Length of the wire, $l = 2 \text{ m}$

Cross-sectional area, $A = 0.5 \text{ mm}^2 = 0.5 \times 10^{-6} \text{ m}^2$

Resistivity of copper, $\rho = 1.7 \times 10^{-8} \Omega\text{-m}$

The resistance R of a wire is given by the formula:

$$R = \rho l / A$$

Step 1: Calculate the initial resistance R_1 and $l = 2 \text{ m}$

$$R_1 = (1.7 \times 10^{-8} \times 2) / (0.5 \times 10^{-6}) = 0.068 \Omega$$

Step 2: Calculate the new resistance R_2 and $l = 4 \text{ m}$ (double length)

$$R_2 = (1.7 \times 10^{-8} \times 4) / (0.5 \times 10^{-6}) = 0.136 \Omega$$

The resistance of the wire when the length is double is 0.136Ω

Verified solution — SQP_2025-26_q34_VI_B

Source Q34 VI B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 2 | Solution crop 1

OR

B.

$$R = \frac{\rho \cdot l}{A}$$

Given:

- Initial length, $l = 2 \text{ m}$
- Cross-sectional area, $A = 0.5 \text{ mm}^2 = 0.5 \times 10^{-6} \text{ m}^2$
- Resistivity of copper, $\rho = 1.7 \times 10^{-8} \Omega \cdot \text{m}$

Step 1: Calculate the initial resistance R_1 and $l = 2 \text{ m}$

$$R_1 = \frac{\rho \cdot l}{A} = \frac{1.7 \times 10^{-8} \Omega \cdot \text{m} \times 2 \text{ m}}{0.5 \times 10^{-6} \text{ m}^2}$$
$$R_1 = \frac{3.4 \times 10^{-8}}{0.5 \times 10^{-6}} \Omega = 0.068 \Omega$$

Step 2: Calculate the new resistance R_2 and $l = 4 \text{ m}$ (double length)

$$R_2 = \frac{\rho \cdot (2l)}{A} = 2 \times R_1 = 2 \times 0.068 \Omega = 0.136 \Omega$$

The resistance of the wire when the length is double is 0.136Ω

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

39. SQP_2025-26_q36 — Source Q36 | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 3

(i) Show that the cross-sectional area of the wire is about $5 \times 10^{-7} \text{ m}^2$.

The cross-sectional area A of a wire with diameter d is given by:

$$A = \pi d^2/4$$

Substitute, $d = 0.80 \times 10^{-3} \text{ m}$:

$$A = \pi(0.80 \times 10^{-3})^2/4$$

$$A = \pi(0.64 \times 10^{-6})/4$$

$$A = (2.01 \times 10^{-6})/4 \approx 5 \times 10^{-7} \text{ m}^2$$

Thus, the cross-sectional area A is approximately $5 \times 10^{-7} \text{ m}^2$.

(ii) To find the length l of the wire, we can use the formula of resistance:

$$R = \rho l/A$$

Rearrange to solve for l :

$$l = RA/\rho$$

Substitute the values:

$$l = (0.12 \times 5 \times 10^{-7})/(1.8 \times 10^{-8})$$

$$l = (6 \times 10^{-8})/(1.8 \times 10^{-8})$$

$$l = 6/1.8 \approx 3.33 \text{ m}$$

The student needs a length of approximately 3.33 m of given copper wire to make a 0.12Ω resistor.

Verified solution — SQP_2025-26_q36

Source Q36 | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 3 | Solution crop 1

36

- (i) Show that the cross-sectional area of the wire is about $5 \times 10^{-7} \text{ m}^2$.
The cross-sectional area A of a wire with diameter d is given by:

$$A = \pi \left(\frac{d}{2} \right)^2$$

Substitute, $d = 0.80 \times 10^{-3} \text{ m}$:

$$A = \pi \left(\frac{0.80 \times 10^{-3}}{2} \right)^2$$

$$A = \pi (0.40 \times 10^{-3})^2$$

$$A = \pi \times (0.16 \times 10^{-6}) \text{ m}^2$$

$$A \approx 3.14 \times 0.16 \times 10^{-6} \text{ m}^2$$

$$A \approx 5.024 \times 10^{-7} \text{ m}^2$$

Thus, the cross-sectional area A is approximately $5 \times 10^{-7} \text{ m}^2$.

- (ii) To find the length l of the wire, we can use the formula of resistance:

$$R = \frac{\rho \cdot l}{A}$$

Rearrange to solve for l :

$$l = \frac{R \cdot A}{\rho}$$

Substitute the values:

$$l = \frac{0.12 \cdot 5 \times 10^{-7}}{1.8 \times 10^{-8}}$$

$$l = \frac{6 \times 10^{-8}}{1.8 \times 10^{-8}}$$

$$l = \frac{6}{1.8} \text{ m}$$

$$l = 3.33 \text{ m}$$

The student needs a length of approximately 3.33 m of given copper wire to make a 0.12Ω resistor.

3

Marking-scheme PDF page(s), one-based: [9]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

40. SQP_2025-26_q39_A — Source Q39 A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

A.

(i) Power across the $100\ \Omega$ resistance = $81\ \text{W}$

$$P = I^2 R = 81\ \text{W}$$

$$\therefore I^2 = 81/100$$

$$\therefore I = \sqrt{(81/100)} = 9/10 = 0.9\ \text{A}$$

(ii) Voltage across the $25\ \Omega$ resistors = $V_2 = IR_{\text{eqv}}$ for the $25\ \Omega$ resistors

$$1/R_{\text{eqv}} = 1/25 + 1/25 = 2/25$$

$$\therefore R_{\text{eqv}} = 25/2 = 12.5\ \Omega$$

$$\therefore V_2 = 0.9\ \text{A} \times 12.5\ \Omega = 11.25\ \text{V}$$

(iii) Voltage across $100\ \Omega = V_{100} = IR = 0.9\ \text{A} \times 100\ \Omega = 90\ \text{V}$

$$\therefore V_1 = 90\ \text{V} + 11.25\ \text{V} = 101.25\ \text{V}$$

Verified solution — SQP_2025-26_q39_A

Source Q39 A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5 | Solution crop 1

39	<u>Students to attempt either option A or B.</u> A. (i) Power across the $100\ \Omega$ resistance = $81\ \text{W}$ $P = I^2 R = 81\ \text{W}$ $\therefore I^2 = \frac{81}{100}$ $\therefore I = \sqrt{\frac{81}{100}} = \frac{9}{10} = 0.9\ \text{A}$	5
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Verified solution — SQP_2025-26_q39_A

Source Q39 A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5 | Solution crop 2

(ii) Voltage across the $25\ \Omega$ resistors $= V_2 = IR_{eqv}$
for the $25\ \Omega$ resistors

$$\frac{1}{R_{eqv}} = \frac{1}{25} + \frac{1}{25} = \frac{2}{25}$$

$$\therefore R_{eqv} = \frac{25}{2} = 12.5\ \Omega$$

$$\therefore V_2 = 0.9\ A \times 12.5\ \Omega = 11.25\ V$$

(iii) Voltage across $100\ \Omega = V_{100} = IR = 0.9\ A \times 100\ \Omega = 90\ V$

$$\therefore V_1 = 90\ V + 11.25\ V = 101.25\ V$$

OR

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

41. SQP_2025-26_q39_B — Source Q39 B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

B.

(i) $P = V^2/R$

$\therefore R = V^2/P = (240 \times 240)/1200 = 48 \Omega$

(ii) For S_1 and S_3 closed

– Current in C

$V = IR$ (Ohm's Law)

$\therefore I = V/R = 240 \text{ V}/48 \Omega = 5 \text{ A}$

– Current in A and B

$V = IR$ (Ohm's Law)

$\therefore I = V/R = 240 \text{ V}/96 \Omega = 2.5 \text{ A}$

(iii) Power across A for S_1, S_2, S_3 closed

$P_A = I^2 R = 5^2 \times 48 = 1200 \text{ W} = 1.2 \text{ KW}$

Verified solution — SQP_2025-26_q39_B

Source Q39 B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5 | Solution crop 1

B.

(i) $P = \frac{V^2}{R}$

$$\therefore R = \frac{V^2}{P} = \frac{240 \times 240}{1200} = 48 \Omega$$

(ii) For S_1 and S_3 closed

– Current in C

$$V = IR \text{ (Ohm's Law)}$$

$$\therefore I = \frac{V}{R} = \frac{240 V}{48 \Omega} = 5 A$$

– Current in A and B

$$V = IR \text{ (Ohm's Law)}$$

$$\therefore I = \frac{V}{R} = \frac{240 V}{96 \Omega} = 2.5 A$$

(iii) Power across A for S_1, S_2, S_3 closed

$$P_A = I^2 R = 5^2 \times 48 = 1200 W = 1.2 KW$$

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

42. SQP_2025-26_q39_VI_A — Source Q39 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

A.

(i) In household circuits, the fuse is connected in series with all appliances to ensure that it can cut off the entire circuit in case of excessive current, preventing hazards like fires or damage. This way, any overload or short circuit causes the fuse to blow, protecting all appliances.

Appliances are connected in parallel to ensure each receives the same voltage from the mains and can operate independently. This setup allows appliances to work simultaneously and efficiently, with each drawing only the current it needs, without affecting others.

Give Data:

Power of heater, $P_{\text{heater}} = 1500 \text{ W}$
Power of Fan, $P_{\text{fan}} = 500 \text{ W}$
Supply Voltage, $V = 220 \text{ V}$
Fuse rating = 10 A

Step 1: Calculate the Current Drawn by Each Appliance

$I = P/V$

1. Current drawn by the heater: $I_{\text{heater}} = 1500/220 = 6.82 \text{ A}$ (rounded to two decimal places)
2. Current drawn by the fan: $I_{\text{fan}} = P_{\text{fan}}/V = 500 \text{ W}/220 \text{ V}$
 $I_{\text{fan}} = 2.27 \text{ A}$ (rounded to two decimal places)

Step 2 : Calculate the total current in the circuit

Since the heater and fan are connected in parallel, the total current I_{total} is the sum of the currents through each appliance:

$I_{\text{total}} = I_{\text{heater}} + I_{\text{fan}} = 6.82 + 2.27 = 9.09 \text{ A}$

Step 3: Compare with the Fuse Rating

The fuse is rated for 10 A, and the total current drawn by the heater and fan together is 9.09 A. Since $9.09 \text{ A} < 10 \text{ A}$, the fuse will not blow and is appropriate for this circuit, as the total current is within the fuse's capacity.

Verified solution — SQP_2025-26_q39_VI_A

Source Q39 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5 | Solution crop 1

For visually impaired students

A.

- (i) In household circuits, the fuse is connected in series with all appliances to ensure that it can cut off the entire circuit in case of excessive current, preventing hazards like fires or damage. This way, any overload or short circuit causes the fuse to blow, protecting all appliances.

Appliances are connected in parallel to ensure each receives the same voltage from the mains and can operate independently. This setup allows appliances to work simultaneously and efficiently, with each drawing only the current it needs, without affecting others.

Give Data:

- Power of heater, $P_{\text{heater}} = 1500 \text{ W}$
- Power of Fan, $P_{\text{fan}} = 500 \text{ W}$

Verified solution — SQP_2025-26_q39_VI_A

Source Q39 VI A | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5 | Solution crop 2

- Supply Voltage, $V = 220 \text{ V}$
- Fuse rating = 10 A

Step 1: Calculate the Current Drawn by Each Appliance

Using the formula = $I = \frac{P}{V}$:

1. Current drawn by the heater:

$$I_{\text{heater}} = \frac{P_{\text{heater}}}{V} = \frac{1500 \text{ W}}{220 \text{ V}}$$

$$I_{\text{heater}} = 6.82 \text{ A (rounded to two decimal places)}$$

2. Current drawn by the fan:

$$I_{\text{fan}} = \frac{P_{\text{fan}}}{V} = \frac{500 \text{ W}}{220 \text{ V}}$$

$$I_{\text{fan}} = 2.27 \text{ A (rounded to two decimal places)}$$

Step 2 : Calculate the total current in the circuit

Since the heater and fan are connected in parallel, the total current I_{total} is the sum of the currents through each appliance:

$$I_{\text{total}} = I_{\text{heater}} + I_{\text{fan}}$$

$$I_{\text{total}} = 6.82 \text{ A} + 2.27 \text{ A}$$

$$I_{\text{total}} = 9.09 \text{ A}$$

Step 3: Compare with the Fuse Rating

The fuse is rated for 10 A , and the total current drawn by the heater and fan together is 9.09 A .

Since $9.09 \text{ A} < 10 \text{ A}$, the fuse will not blow and is appropriate for this circuit, as the total current is within the fuse's capacity.

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

43. SQP_2025-26_q39_VI_B — Source Q39 VI B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5

B.

Given data:

Resistor $R_1 = 6\ \Omega$

Resistor $R_2 = 12\ \Omega$

Voltage $V = 24\text{ V}$

Time $t = 5\text{ Minutes} = 5 \times 60 = 300\text{ seconds}$

Step 1: Calculate the Current through each Resistor

Since the resistors are connected in parallel, the voltage across each resistor is the same as the battery voltage, $V = 24\text{ V}$.

Using Ohm's Law,

Current through R_1 : $I_1 = V/R_1 = 24/6 = 4\text{ A}$

Current through R_2 : $I_2 = V/R_2 = 24/12 = 2\text{ A}$

Step 2: Calculate the Heat Generated in Each Resistor

Using Joule's Law of Heating, $H = I^2Rt$:

Heat generate in R_1 : $H_1 = 4^2 \times 6 \times 300 = 28800\text{ J}$

Heat generate in R_2 : $H_2 = 2^2 \times 12 \times 300 = 14400\text{ J}$

Total Heat Generated H : $H = H_1 + H_2 = 28800 + 14400 = 43200\text{ J}$

So, the total heat generated in both resistor is 43200 J .

Step 3 : Determine if each Resistor is safe

The power dissipated by each resistor can be calculated using $P = V \times I$

Power dissipated by R_1 : $P_1 = 24 \times 4 = 96\text{ W}$

Power dissipated by R_2 : $P_2 = 24 \times 2 = 48\text{ W}$

Given that the power rating of each resistor is 100 W :

R_1 is operating at 96 W , which is within the 100 W limit. Hence, it is safe.

R_2 is operating at 48 W , which is also within the 100 W limit. Hence, it is safe.

Verified solution — SQP_2025-26_q39_VI_B

Source Q39 VI B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5 | Solution crop 1

B.

Given data:

- Resistor $R_1 = 6 \, \Omega$
- Resistor $R_2 = 12 \, \Omega$
- Voltage $V = 24 \, \text{V}$
- Time $t = 5 \text{ Minutes} = 5 \times 60 = 300 \text{ seconds}$

Step 1: Calculate the Current through each Resistor

Since the resistors are connected in parallel, the voltage across each resistor is the same as the battery voltage, $V = 24 \, \text{V}$.

Using Ohm's Law, $I = \frac{V}{R}$:

Verified solution — SQP_2025-26_q39_VI_B

Source Q39 VI B | Year: 2025-26 | Session: 2025-26 | Code: Not recorded | Marks: 5 | Solution crop 2

Current through R_1

$$I_1 = \frac{V}{R_1} = \frac{24\text{ V}}{6\ \Omega} = 4\text{ A}$$

Current through R_2 :

$$I_2 = \frac{V}{R_2} = \frac{24\text{ V}}{12\ \Omega} = 2\text{ A}$$

Step 2: Calculate the Heat Generated in Each Resistor

Using Joule's Law of Heating, $H = I^2 R t$:

- Heat generate in R_1 :

$$H_1 = I_1^2 \times R_1 \times t = (4\text{ A})^2 \times 6\ \Omega \times 300\text{ s}$$

$$H_1 = 16 \times 6 \times 300 = 28800\text{ J}$$

- Heat generate in R_2 :

$$H_2 = I_2^2 \times R_2 \times t = (2\text{ A})^2 \times 12\ \Omega \times 300\text{ s}$$

$$H_2 = 4 \times 12 \times 300 = 14400\text{ J}$$

- Total Heat Generated H :

$$H_{\text{total}} = H_1 + H_2 = 28800\text{ J} + 14400\text{ J} = 43200\text{ J}$$

So, the total heat generated in both resistor is **43200 J**.

Step 3 : Determine if each Resistor is safe

The power dissipated by each resistor can be calculated using $P = V \times I$

- Power dissipated by R_1 :

$$P_1 = V \times I_1 = 24\text{ V} \times 4\text{ A} = 96\text{ W}$$

- Power dissipated by R_2 :

$$P_2 = V \times I_2 = 24\text{ V} \times 2\text{ A} = 48\text{ W}$$

Given that the power rating of each resistor is 100 W:

- R_1 is operating at 96 W, which is within the 100 W limit. Hence, it is safe.
- R_2 is operating at 48 W, which is also within the 100 W limit. Hence, it is safe.

Marking-scheme PDF page(s), one-based: [14, 15]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

**44. SQP_2026-27_q31 — Source Q31 | Year: 2026-27 | Session: 2026-27 |
Code: Not recorded | Marks: 1**

C. 4.0 A

Verified solution — SQP_2026-27_q31

Source Q31 | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 1 | Solution crop 1

31 | C. 4.0 A

1 |

Marking-scheme PDF page(s), one-based: [7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

45. SQP_2026-27_q34_A — Source Q34 A | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 2

A.

I. Joule's law of heating: The heat produced in a resistor is

- (i) directly proportional to the square of current for a given resistance,
- (ii) directly proportional to resistance for a given current, and
- (iii) directly proportional to the time for which the current flows through the resistor.

$$H = I^2Rt,$$

where 'I' is current and 'R' is resistance. (1)

II. If resistance is doubled: For the same current I, heat H becomes

$$H = I^2(2R)t = 2I^2Rt = 2H,$$

i.e. heat doubles. (1)

Verified solution — SQP_2026-27_q34_A

Source Q34 A | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 2 | Solution crop 1

34	<u>Student to attempt either A or B.</u> A. I. Joule's law of heating: The heat produced in a resistor is (i) directly proportional to the square of current for a given resistance, (ii) directly proportional to resistance for a given current, and (iii) directly proportional to the time for which the current flows through the resistor. $H = I^2Rt$, where 'I' is current and 'R' is resistance. (1) II. If resistance is doubled: For the same current I, heat H becomes $H = I^2(2R)t = 2I^2Rt = 2H,$ i.e. heat doubles. (1)	2
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Marking-scheme PDF page(s), one-based: [7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

46. SQP_2026-27_q34_B — Source Q34 B | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 2

B.

I. Definition: The resistivity (ρ) of a material is defined as the resistance of a wire having unit cross-sectional area and unit length

$R = \rho L/A$, where R is resistance of a wire of length L and cross-sectional area A (1)

II. Intrinsic property justification: Resistivity depends only on material (not on dimensions). Example: Two copper wires of different lengths and thick-nesses yield the same ρ when computed using $R \cdot A/L$. (1)

Verified solution — SQP_2026-27_q34_B

Source Q34 B | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 2 | Solution crop 1

B.

- I. Definition: The resistivity (ρ) of a material is defined as the resistance of a wire having unit cross-sectional area and unit length $R = \frac{\rho L}{A}$, where R is resistance of a wire of length L and cross-sectional area A (1)
- II. Intrinsic property justification: Resistivity depends only on material (not on dimensions). Example: Two copper wires of different lengths and thick-nesses yield the same ρ when computed using $R \cdot A/L$. (1)

Marking-scheme PDF page(s), one-based: [7]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

47. SQP_2026-27_q37_main — Source Q37 main | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 3

A.

1 Ω and 1 Ω are in series and the combination is in parallel with 2 Ω

$$1/R = 1/2 + 1/2 \quad R_{AB} = 1 \Omega$$

$$1/R = 1/3 + 1/3 \quad R = 1.5 \Omega$$

5 Ω , 1 Ω and 1.5 Ω are in series: $R_s = 5 + 1 + 1.5$

$$R_s = 7.5 \Omega$$

$$V = IR$$

$$15 V = I \times 7.5$$

$$I = 2A \quad (2)$$

B. $R_{AB} = 1 \Omega$, net current between points A and B is 2A

$$V_{AB} = I_{AB} \times R_{AB}$$

$$V_{AB} = 2 \times 1 = 2V \quad (1)$$

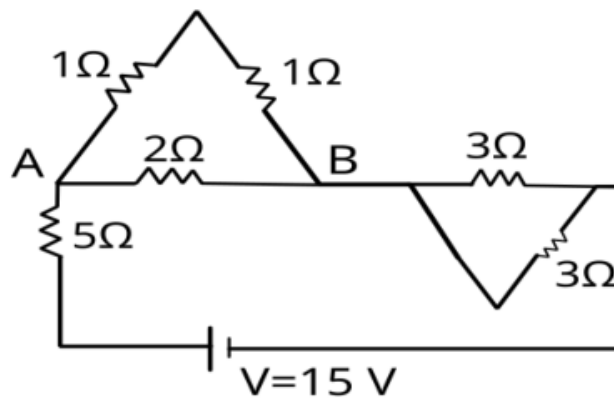
Verified solution — SQP_2026-27_q37_main

Source Q37 main | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 3 | Solution crop 1

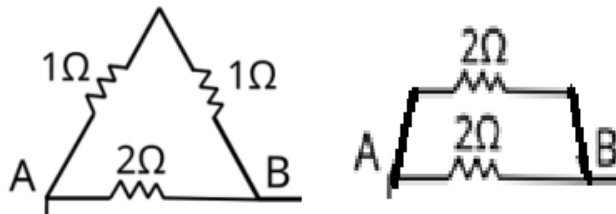
37

A.

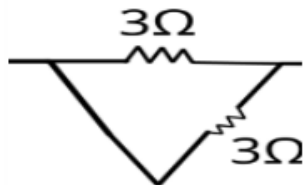
3



1 Ω and 1 Ω are in series and the combination is in parallel with 2 Ω



$$\frac{1}{R} = \frac{1}{2} + \frac{1}{2} \quad R_{AB} = 1 \Omega$$



$$\frac{1}{R} = \frac{1}{3} + \frac{1}{3} \quad R = 1.5 \Omega$$

5 Ω , 1 Ω and 1.5 Ω are in series: $R_s = 5 + 1 + 1.5$
 $R_s = 7.5 \Omega$

$$V = I R$$

$$15 \text{ V} = I * 7.5$$

$$I = 2 \text{ A}$$

(2)

B. $R_{AB} = 1 \Omega$, net current between points A and B is 2A

$$V_{AB} = I_{AB} * R_{AB}$$

$$V_{AB} = 2 * 1 = 2 \text{ V}$$

(1)

Marking-scheme PDF page(s), one-based: [9]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

48. SQP_2026-27_q37_VI — Source Q37 VI | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 3

A. Total Resistance:

$$R_{\text{total}} = 2 + 3 + 5 = 10 \, \Omega \, (1)$$

B. Total current:

$$I = V/R_{\text{total}} = 15/10 = 1.5 \, \text{A}. \, (1)$$

C. Voltage drop across the $3 \, \Omega$ resistor:

$$V_{3\Omega} = I \times 3 = 1.5 \times 3 = 4.5 \, \text{V} \, (1)$$

Verified solution — SQP_2026-27_q37_VI

Source Q37 VI | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 3 | Solution crop 1

For visually impaired students

A. Total Resistance:

$$R_{\text{total}} = 2 + 3 + 5 = 10 \, \Omega$$

B. Total current:

(1)

Verified solution — SQP_2026-27_q37_VI

Source Q37 VI | Year: 2026-27 | Session: 2026-27 | Code: Not recorded | Marks: 3 | Solution crop 2

	$I = \frac{V}{R_{\text{total}}} = \frac{15}{10} = 1.5 \text{ A.}$	(1)
	C. Voltage drop across the 3Ω resistor: $V_{3 \Omega} = I \times 3 = 1.5 \times 3 = 4.5 \text{ V}$	(1)

Marking-scheme PDF page(s), one-based: [9, 10]

Answer verification note: Official MS answer checked in batch review and final answer-text audit. Explicit minor source corrections remain disclosed; source evidence retained.

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

Per-question exact source appendix

Question labels and provenance below are source-record values, not inferred from filenames. Physical source PDF page numbers are one-based. Missing metadata is explicitly marked Not recorded; no source_id or paper code is synthesized.

1. cbse_sqp_2017-18_electricity_q7

Source ID: src-0c6f56c06a42

Source title: Sample Question Paper 2017-18 — Science — Class - X

Exam type: SQP

Year: 2017-18

Session: 2017-18

Paper code: Not recorded

Set: Not recorded

Original question label: 7

Original section: SECTION - A

Source marks (nonadditive): 3

Parent question label: 7

Alternative: 7

Choice group: 2017-18_Q7

Exam mark group: 2017-18_Q7

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [1]

question_crop_regions: [{"bbox": [72, 595, 541, 672], "page": 1}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Question asks the candidate to draw a circuit; no diagram is supplied in the SQP. Original wording "Each of them dissipate maximum energy" retained. In the connected circuit the 4Ω resistor limits the total current to 2 A; each 8Ω resistor then dissipates 8 W, not 16 W. The official answer follows this limiting-current interpretation. Final audit replaced clipped original crop; complete final "resistors." line visually confirmed.

pdf_file: assets/research/3f5dbd20e94bd0e4ae5f.pdf

Source raster asset(s): ["assets/research/9132fd5467ea364b896d.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

2. cbse_sqp_2017-18_electricity_q26

Source ID: src-0c6f56c06a42

Source title: Sample Question Paper 2017-18 — Science — Class - X

Exam type: SQP

Year: 2017-18

Session: 2017-18

Paper code: Not recorded

Set: Not recorded

Original question label: 26

Original section: SECTION - B

Source marks (nonadditive): 2

Parent question label: 26

Alternative: 26

Choice group: 2017-18_Q26

Exam mark group: 2017-18_Q26

Marks include choice: True

Source verified: True

Answer verified: False

Source PDF page(s), one-based: [3, 4]

question_crop_regions: [{"bbox": [74, 674, 539, 752], "page": 3}, {"bbox": [101, 61, 516, 84], "page": 4}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: source_ready_answer_hold

notes: Practical-skills question; continues from SQP page 3 to page 4. OR alternative stored separately. Official MS states $3.2\ \Omega$ and does not supply the plotted graph; data ratios are approximate (about $3.2\text{--}3.4\ \Omega$). The official worked slope uses graph coordinates not listed in the table; preserved rather than silently corrected. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official slope solution captured, but required plotted graph absent from official MS; not a complete publishable model answer. Consult the linked original marking scheme with this limitation in mind.

pdf_file: assets/research/3f5dbd20e94bd0e4ae5f.pdf

Source raster asset(s): ["assets/research/397d1493244de850dc49.png", "assets/research/893648996e6e757d73c8.png"]

Official source: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

3. cbse_sqp_2017-18_electricity_q26_OR

Source ID: src-0c6f56c06a42

Source title: Sample Question Paper 2017-18 — Science — Class - X

Exam type: SQP

Year: 2017-18

Session: 2017-18

Paper code: Not recorded

Set: Not recorded

Original question label: 26 OR

Original section: SECTION – B

Source marks (nonadditive): 2

Parent question label: 26

Alternative: 26 OR

Choice group: 2017-18_Q26

Exam mark group: 2017-18_Q26

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

question_crop_regions: [{"bbox": [101, 88, 516, 139], "page": 4}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Separate internal OR alternative to Q26; practical-skills question. The printed number 26 and 2-mark allocation occur on the previous page for the shared question slot.

pdf_file: assets/research/3f5dbd20e94bd0e4ae5f.pdf

Source raster asset(s): ["assets/research/291dd44d213b557d5ba5.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

4. cbse_sqp_2018-19_electricity_q12

Source ID: src-3896b57b200e

Source title: Class-X Science-086 SAMPLE QUESTION PAPER-19

Exam type: SQP

Year: 2018-19

Session: 2018-19

Paper code: Not recorded

Set: Not recorded

Original question label: 12

Original section: SECTION C

Source marks (nonadditive): 3

Parent question label: 12

Alternative: 12

Choice group: 2018-19_Q12

Exam mark group: 2018-19_Q12

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [2]

question_crop_regions: [{"bbox": [41, 597, 580, 631], "page": 2}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

included_subparts: ["(i) in series", "(ii) in parallel"]

notes: Included despite mains-supply context: self-contained resistance, power and current calculation, not domestic-wiring knowledge.

pdf_file: assets/research/2dbea9870ad27ab587b6.pdf

Source raster asset(s): ["assets/research/72cffce3dabc56fbdeda.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

Official marking scheme:

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

5. cbse_sqp_2018-19_electricity_q13

Source ID: src-3896b57b200e

Source title: Class-X Science-086 SAMPLE QUESTION PAPER-19

Exam type: SQP

Year: 2018-19

Session: 2018-19

Paper code: Not recorded

Set: Not recorded

Original question label: 13

Original section: SECTION C

Source marks (nonadditive): 3

Parent question label: 13

Alternative: 13

Choice group: 2018-19_Q13

Exam mark group: 2018-19_Q13

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [3]

question_crop_regions: [{"bbox": [41, 70, 580, 219], "page": 3}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Diagram labels verified visually: (a) L and A; (b) 3L and A/3; (c) L/3 and 3A. The full question crop preserves a stray printed "3" below the figures as present in the source.

pdf_file: assets/research/2dbea9870ad27ab587b6.pdf

Source raster asset(s): ["assets/research/42eb8dcffafca811ce1.png", "assets/research/ae6b9a10898108d91d59.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

Official marking scheme:

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

6. cbse_sqp_2018-19_electricity_q26

Source ID: src-3896b57b200e

Source title: Class-X Science-086 SAMPLE QUESTION PAPER-19

Exam type: SQP

Year: 2018-19

Session: 2018-19

Paper code: Not recorded

Set: Not recorded

Original question label: 26

Original section: SECTION E

Source marks (nonadditive): 2

Parent question label: 26

Alternative: 26

Choice group: 2018-19_Q26

Exam mark group: 2018-19_Q26

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [5]

question_crop_regions: [{"bbox": [41, 555, 580, 600], "page": 5}, {"bbox": [89, 118, 505, 244], "page": 5}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Practical-skills question. Official SQP layout error: the ammeter/voltmeter image appears beside chemistry Q22 near the top of the SAME PDF page 5, not alongside Q26. The matching meters and MS establish the intended association. Evidence array puts Q26 text first and displaced diagram second. No image was redrawn.

pdf_file: assets/research/2dbea9870ad27ab587b6.pdf

Source raster asset(s): ["assets/research/96a7fabe893320ee944c.png", "assets/research/e71b4ebd5bae14822c41.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

Official marking scheme:

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

7. cbse_sqp_2019-20_electricity_q6

Source ID: src-19fa63b41ec6

Source title: Class-X Science-086 SAMPLE QUESTION PAPER 2019-20

Exam type: SQP

Year: 2019-20

Session: 2019-20

Paper code: Not recorded

Set: Not recorded

Original question label: 6

Original section: SECTION A

Source marks (nonadditive): 1

Parent question label: 6

Alternative: 6

Choice group: 2019-20_Q6

Exam mark group: 2019-20_Q6

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

question_crop_regions: [{"bbox": [65, 96, 540, 205], "page": 4}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Original spelling "resister" retained. Q6 mark is printed on the preceding page (page 3); Section A instructions and the MS both confirm 1 mark.

pdf_file: assets/research/4f627aa1ad5077856e8d.pdf

Source raster asset(s): ["assets/research/1e17848ea97ced19fb0b.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

8. cbse_sqp_2019-20_electricity_q7

Source ID: src-19fa63b41ec6

Source title: Class-X Science-086 SAMPLE QUESTION PAPER 2019-20

Exam type: SQP

Year: 2019-20

Session: 2019-20

Paper code: Not recorded

Set: Not recorded

Original question label: 7

Original section: SECTION A

Source marks (nonadditive): 1

Parent question label: 7

Alternative: 7

Choice group: 2019-20_Q7

Exam mark group: 2019-20_Q7

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

question_crop_regions: [{"bbox": [65, 207, 540, 301], "page": 4}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Not recorded

pdf_file: assets/research/4f627aa1ad5077856e8d.pdf

Source raster asset(s): ["assets/research/4a5a5f41672fe38acb5b.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

9. cbse_sqp_2019-20_electricity_q29

Source ID: src-19fa63b41ec6

Source title: Class-X Science-086 SAMPLE QUESTION PAPER 2019-20

Exam type: SQP

Year: 2019-20

Session: 2019-20

Paper code: Not recorded

Set: Not recorded

Original question label: 29

Original section: SECTION C

Source marks (nonadditive): 5

Parent question label: 29

Alternative: 29

Choice group: 2019-20_Q29

Exam mark group: 2019-20_Q29

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [9]

question_crop_regions: [{"bbox": [64, 188, 526, 328], "page": 9}]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

included_subparts: ["(i)", "(ii)(a)", "(ii)(b)", "(iii)"]

notes: All dependent subparts retained in one 5-mark record, including both (ii)(a) and (ii)(b). Marking-scheme answer spans PDF pages 6–7.

pdf_file: assets/research/4f627aa1ad5077856e8d.pdf

Source raster asset(s): ["assets/research/eaf3ddd57a4e63ec45d3.png"]

answer_transcription: Faithful summary of official marking scheme; exact source image retained.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

10. SQP-2019-20-Q14

Source ID: src-19fa63b41ec6

Source title: Science-086 Sample Question Paper 2019-20 Class-X

Exam type: SQP

Year: 2019-20

Session: 2019-20

Paper code: Not recorded

Set: Not recorded

Original question label: 14

Original section: SECTION A

Source marks (nonadditive): 1

Parent question label: 14

Alternative: 14

Choice group: 2019-20_Q14

Exam mark group: 2019-20_Q14

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [5, 6]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Fuse series placement and melting from excess current are explicitly NCERT Electricity section 11.7.1 (printed p190, PDF p20). Included after independent audit overturned the batch reviewer's domestic-circuit exclusion. Shared assertion-reason choices occur on preceding PDF page 5.

pdf_file: assets/research/4f627aa1ad5077856e8d.pdf

Source raster asset(s): ["assets/research/3f7478207b39964e13b3.png", "assets/research/db446bcbde928527a58e.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

11. sqp_2020-21_q9_main

Source ID: src-ef1e3e045e9e

Source title: CBSE Class X Science (086) Sample Question Paper 2020-21

Exam type: SQP

Year: 2020-21

Session: 2020-21

Paper code: Not recorded

Set: Not recorded

Original question label: 9

Original section: Section A

Source marks (nonadditive): 1

Parent question label: 9

Alternative: 9

Choice group: 2020-21_Q9

Exam mark group: 2020-21_Q9

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [2]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

choice_limit: 1

notes: One-mark very short answer, not MCQ. Main and OR alternatives share one exam mark; do not add their marks together.

pdf_file: assets/research/b7be8cafa2de88a7e7a8.pdf

Source raster asset(s): ["assets/research/52891be9e4f2da1108fa.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

12. sqp_2020-21_q9_OR

Source ID: src-ef1e3e045e9e

Source title: CBSE Class X Science (086) Sample Question Paper 2020-21

Exam type: SQP

Year: 2020-21

Session: 2020-21

Paper code: Not recorded

Set: Not recorded

Original question label: 9

Original section: Section A

Source marks (nonadditive): 1

Parent question label: 9

Alternative: 9

Choice group: 2020-21_Q9

Exam mark group: 2020-21_Q9

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [2]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

choice_limit: 1

notes: One-mark very short answer, not MCQ. OR alternative to Q9 main; one shared exam mark.

pdf_file: assets/research/b7be8cafa2de88a7e7a8.pdf

Source raster asset(s): ["assets/research/42720098337bd4c9dbbe.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

13. sqp_2020-21_q26

Source ID: src-ef1e3e045e9e

Source title: CBSE Class X Science (086) Sample Question Paper 2020-21

Exam type: SQP

Year: 2020-21

Session: 2020-21

Paper code: Not recorded

Set: Not recorded

Original question label: 26

Original section: Section B

Source marks (nonadditive): 2

Parent question label: 26

Alternative: 26

Choice group: 2020-21_Q26

Exam mark group: 2020-21_Q26

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [12]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Not recorded

pdf_file: assets/research/b7be8cafa2de88a7e7a8.pdf

Source raster asset(s): ["assets/research/4a08b523353b5d275bb1.png",
"assets/research/7129bcaaaa6f1ee587dc.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

14. sqp_2020-21_q36_main

Source ID: src-ef1e3e045e9e

Source title: CBSE Class X Science (086) Sample Question Paper 2020-21

Exam type: SQP

Year: 2020-21

Session: 2020-21

Paper code: Not recorded

Set: Not recorded

Original question label: 36

Original section: Section D

Source marks (nonadditive): 5

Parent question label: 36

Alternative: 36

Choice group: 2020-21_Q36

Exam mark group: 2020-21_Q36

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [14]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

choice_limit: 1

notes: All four subparts retained. The OR alternative on pp.14–15 concerns magnetic fields and the right-hand thumb rule and is excluded. Source MS p.8 prints " $= 60/15 \text{ A} = 4\text{A}$ "; the resistance unit is corrected explicitly to Ω , not silently attributed to the source.

pdf_file: assets/research/b7be8cafa2de88a7e7a8.pdf

Source raster asset(s): ["assets/research/87a627e14877e30edf2b.png",
"assets/research/93c985cc23d73326d2d9.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

15. sqp_2021-22_term2_q11

Source ID: src-3dd66b4fc7f1

Source title: CBSE Class X Science (086) Sample Question Paper 2021-22 — Term 2

Exam type: SQP

Year: 2021-22

Session: 2021-22

Paper code: Not recorded

Set: Not recorded

Original question label: 11

Original section: Section B

Source marks (nonadditive): 3

Parent question label: 11

Alternative: 11

Choice group: 2021-22-T2_Q11

Exam mark group: 2021-22-T2_Q11

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

subpart_marks: {"a": 1, "b": 2}

notes: Both compulsory subparts retained in one three-mark record. MS answer spans pp.3–4.

pdf_file: assets/research/c1e69bd521d5de2220f0.pdf

Source raster asset(s): ["assets/research/bab15ec8b02072aa30b0.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-SQP_Term2.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-MS_Term2.pdf

16. sqp_2021-22_term2_q12_main

Source ID: src-3dd66b4fc7f1

Source title: CBSE Class X Science (086) Sample Question Paper 2021-22 — Term 2

Exam type: SQP

Year: 2021-22

Session: 2021-22

Paper code: Not recorded

Set: Not recorded

Original question label: 12

Original section: Section B

Source marks (nonadditive): 3

Parent question label: 12

Alternative: 12

Choice group: 2021-22-T2_Q12

Exam mark group: 2021-22-T2_Q12

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

choice_limit: 1

notes: Main and OR are separate three-mark alternatives sharing a single three-mark exam slot. MS incorrectly calls the $5\ \Omega$, $10\ \Omega$ and R_1 block “in series”, although its reciprocal formula and source circuit show parallel; MS also prints $R_{eq} = 15\ A$ instead of $15\ \Omega$. These source errors are explicitly corrected in answer.

pdf_file: assets/research/c1e69bd521d5de2220f0.pdf

Source raster asset(s): ["assets/research/3eb95886c16de4c96741.png",
"assets/research/c028200f8ca6a07212ea.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-SQP_Term2.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-MS_Term2.pdf

17. sqp_2021-22_term2_q12_OR

Source ID: src-3dd66b4fc7f1

Source title: CBSE Class X Science (086) Sample Question Paper 2021-22 — Term 2

Exam type: SQP

Year: 2021-22

Session: 2021-22

Paper code: Not recorded

Set: Not recorded

Original question label: 12

Original section: Section B

Source marks (nonadditive): 3

Parent question label: 12

Alternative: 12

Choice group: 2021-22-T2_Q12

Exam mark group: 2021-22-T2_Q12

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

choice_limit: 1

notes: OR alternative with its own complete circuit and stem. Official solution is an embedded image on MS p.4 and was visually transcribed. Main and OR share three exam marks.

pdf_file: assets/research/c1e69bd521d5de2220f0.pdf

Source raster asset(s): ["assets/research/aaa7acc458025ae4d407.png", "assets/research/8f730b36669fc856adfa.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-SQP_Term2.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-MS_Term2.pdf

18. sqp_2022-23_q13

Source ID: src-7396f04b9b1c

Source title: CBSE Class X Science (086) Sample Question Paper 2022-23

Exam type: SQP

Year: 2022-23

Session: 2022-23

Paper code: Not recorded

Set: Not recorded

Original question label: 13

Original section: Section A

Source marks (nonadditive): 1

Parent question label: 13

Alternative: 13

Choice group: 2022-23_Q13

Exam mark group: 2022-23_Q13

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Not recorded

pdf_file: assets/research/6834135042a4c9866d8e.pdf

Source raster asset(s): ["assets/research/215766334fe1f84dc36b.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

19. sqp_2022-23_q15

Source ID: src-7396f04b9b1c

Source title: CBSE Class X Science (086) Sample Question Paper 2022-23

Exam type: SQP

Year: 2022-23

Session: 2022-23

Paper code: Not recorded

Set: Not recorded

Original question label: 15

Original section: Section A

Source marks (nonadditive): 1

Parent question label: 15

Alternative: 15

Choice group: 2022-23_Q15

Exam mark group: 2022-23_Q15

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [5]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Not recorded

pdf_file: assets/research/6834135042a4c9866d8e.pdf

Source raster asset(s): ["assets/research/15715be9788a3274d549.png",
"assets/research/2eb8ad843f0beb9a883b.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

20. sqp_2022-23_q36

Source ID: src-7396f04b9b1c

Source title: CBSE Class X Science (086) Sample Question Paper 2022-23

Exam type: SQP

Year: 2022-23

Session: 2022-23

Paper code: Not recorded

Set: Not recorded

Original question label: 36

Original section: Section D

Source marks (nonadditive): 5

Parent question label: 36

Alternative: 36

Choice group: 2022-23_Q36

Exam mark group: 2022-23_Q36

Marks include choice: True

Source verified: True

Answer verified: False

Source PDF page(s), one-based: [9]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: source_ready_answer_hold

marks_status: 5 explicitly printed for complete question; c/d split unavailable in MS.

original_question_marks: 5

included_subparts: ["a", "b", "c", "d"]

excluded_subparts: []

subpart_marks: {"a": 1, "b": 2, "c": null, "d": null}

notes: Full Q36(a-d) retained after NCERT-based chapter audit: parallel circuits and resistances are Electricity, and protective fuse function is explicitly NCERT Electricity section 11.7.1 (printed p190). Printed parent total = 5 marks. Official MS p6 supplies (a) and (b), but omits (c) and (d). Do not infer subpart mark allocations for those omissions. Answer (c)=11 A is independently derived, not an official MS answer. Answer (d) is not supplied here as an official answer. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official MS omits c,d; independent c derivation labelled; complete answer not verified. Consult the linked original marking scheme with this limitation in mind.

pdf_file: assets/research/6834135042a4c9866d8e.pdf

Source raster asset(s): ["assets/research/d3c5899332e8c8dc0654.png", "assets/research/be33989c028e9c57cc40.png"]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

21. SQP_2023-24_q25_main

Source ID: src-3a7e4a2d99b8

Source title: CBSE Class X Science (086) Sample Question Paper 2023-24 (English)

Exam type: SQP

Year: 2023-24

Session: 2023-24

Paper code: Not recorded

Set: Not recorded

Original question label: 25 main

Original section: Section-B

Source marks (nonadditive): 2

Parent question label: 25

Alternative: 25 main

Choice group: 2023-24_Q25

Exam mark group: 2023-24_Q25

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [4]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Main alternative only. OR concerns magnetic field patterns and is excluded. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ebba888ab63fb8276778.pdf

Source raster asset(s): ["assets/research/17bef497a61bc89f5443.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2023-24_q25_main_MS_1.png", "page": 2, "pdf_file": "research/sqp/sources/2023-24_Science-MS.pdf", "rect": [45.0, 73.5, 550.0, 167]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

22. SQP_2023-24_q32

Source ID: src-3a7e4a2d99b8

Source title: CBSE Class X Science (086) Sample Question Paper 2023-24 (English)

Exam type: SQP

Year: 2023-24

Session: 2023-24

Paper code: Not recorded

Set: Not recorded

Original question label: 32

Original section: Section-C

Source marks (nonadditive): 3

Parent question label: 32

Alternative: 32

Choice group: 2023-24_Q32

Exam mark group: 2023-24_Q32

Marks include choice: True

Source verified: True

Answer verified: False

Source PDF page(s), one-based: [5]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: source_ready_answer_hold

notes: Official MS prints $H = 1^2Rt$ (digit 1, visually verified), evidently a typo for $H = I^2Rt$. Printed typo preserved in answer. Marks 2+1. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official answer retains erroneous digit 1 in $H = 1^2Rt$; correct formula explained in notes. Consult the linked original marking scheme with this limitation in mind.

pdf_file: assets/research/ebba888ab63fb8276778.pdf

Source raster asset(s): ["assets/research/3f99dba9221dd54d869a.png"]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

23. SQP_2023-24_q39_iii

Source ID: src-3a7e4a2d99b8

Source title: CBSE Class X Science (086) Sample Question Paper 2023-24 (English)

Exam type: SQP

Year: 2023-24

Session: 2023-24

Paper code: Not recorded

Set: Not recorded

Original question label: 39 iii

Original section: Section-E

Source marks (nonadditive): 4

Parent question label: 39

Alternative: 39 iii

Choice group: 2023-24_Q39

Exam mark group: 2023-24_Q39

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [8]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Includes common (i), (ii) and main (iii); 1+1+2 marks. Original scheme mislabels second line of (i) as (a). Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ebba888ab63fb8276778.pdf

Source raster asset(s): ["assets/research/e0f4305d1e2fbcad0a5b.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2023-24_q39_iii_MS_1.png", "page": 6, "pdf_file": "research/sqp/sources/2023-24_Science-MS.pdf", "rect": [45.0, 335.0, 550.0, 525.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

24. SQP_2023-24_q39_OR_iv

Source ID: src-3a7e4a2d99b8

Source title: CBSE Class X Science (086) Sample Question Paper 2023-24 (English)

Exam type: SQP

Year: 2023-24

Session: 2023-24

Paper code: Not recorded

Set: Not recorded

Original question label: 39 OR iv

Original section: Section-E

Source marks (nonadditive): 4

Parent question label: 39

Alternative: 39 OR iv

Choice group: 2023-24_Q39

Exam mark group: 2023-24_Q39

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [8]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Includes common (i), (ii) and OR (iv); 1+1+2 marks. Original scheme mislabels second line of (i) as (a). Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ebba888ab63fb8276778.pdf

Source raster asset(s): ["assets/research/62bf5ddef468e04d42cc.png", "assets/research/1eda62d03f97318b251f.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2023-24_q39_OR_iv_MS_1.png", "page": 6, "pdf_file": "research/sqp/sources/2023-24_Science-MS.pdf", "rect": [45.0, 335.0, 550.0, 386.0]}, {"file": "research/sqp/evidence/recent_2023-24_q39_OR_iv_MS_2.png", "page": 6, "pdf_file": "research/sqp/sources/2023-24_Science-MS.pdf", "rect": [45.0, 525, 550.0, 633]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

25. SQP-2023-24-Q33-ii

Source ID: src-3a7e4a2d99b8

Source title: Science (086) Sample Question Paper Class X (2023-24)

Exam type: SQP

Year: 2023-24

Session: 2023-24

Paper code: Not recorded

Set: Not recorded

Original question label: 33(ii)

Original section: Section-C

Source marks (nonadditive): 1

Parent question label: 33

Alternative: 33(ii)

Choice group: 2023-24_Q33

Exam mark group: 2023-24_Q33

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [6]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Self-contained Electricity subpart from mixed parent Q33. Q33(i) concerns earthing and is excluded. Fuse protection and melting are explicitly Electricity, NCERT section 11.7.1, printed p190 / PDF p20. One-mark VSA requires

short1 type extension; not MCQ. Parent question is 3 marks; this subpart is 1 mark.

pdf_file: assets/research/ebba888ab63fb8276778.pdf

Source raster asset(s): ["assets/research/77227b54f394dabee412.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

26. SQP_2024-25_q25_A

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 25 A

Original section: Section-B

Source marks (nonadditive): 2

Parent question label: 25

Alternative: 25 A

Choice group: 2024-25_Q25

Exam mark group: 2024-25_Q25

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [6]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/0225289ffd5f7b7cc446.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2024-25_q25_A_MS_1.png", "page": 3, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 53.0, 550.0, 138.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

27. SQP_2024-25_q25_B

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 25 B

Original section: Section-B

Source marks (nonadditive): 2

Parent question label: 25

Alternative: 25 B

Choice group: 2024-25_Q25

Exam mark group: 2024-25_Q25

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [6, 7]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Diagram supplies $X = R$ and $Y = 2R$; continuation on next page retained. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/6a6ec1eda960f8d86c18.png", "assets/research/c2368fa653a77130dc62.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2024-25_q25_B_MS_1.png", "page": 3, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 158, 550.0, 264]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

28. SQP_2024-25_q25_VI_A

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 25 VI A

Original section: Section-B

Source marks (nonadditive): 2

Parent question label: 25

Alternative: 25 VI A

Choice group: 2024-25_Q25

Exam mark group: 2024-25_Q25

Marks include choice: True

Source verified: True

Answer verified: False

Source PDF page(s), one-based: [7]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: source_ready_answer_hold

notes: Official MS arithmetic error retained: $1/12 = 3/36$, not $2/36$. Independently calculated correct minimum = $36/25 = 1.44 \Omega$; this correction is not substituted into the official answer. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official arithmetic answer 1.5 ohm is erroneous; correct 1.44 ohm documented separately. Consult the linked original marking scheme with this limitation in mind.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/a182e9c16e801d9d25f7.png"]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

29. SQP_2024-25_q25_VI_B

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 25 VI B

Original section: Section-B

Source marks (nonadditive): 2

Parent question label: 25

Alternative: 25 VI B

Choice group: 2024-25_Q25

Exam mark group: 2024-25_Q25

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [7]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Borderline household safety context retained by explicit scope instruction: electric-power/current calculation and fuse rating. Original MS wording "with is" retained. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/ddd1c2537a2ff75d83ae.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2024-25_q25_VI_B_MS_1.png", "page": 3, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 541.0, 550.0, 650.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

30. SQP_2024-25_q32

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 32

Original section: Section-C

Source marks (nonadditive): 3

Parent question label: 32

Alternative: 32

Choice group: 2024-25_Q32

Exam mark group: 2024-25_Q32

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [9]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: The intermediate MS proportionality is printed $R \propto I/A$; retained rather than silently changing it to $R \propto 1/A$. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/f51d3e028e24147adb27.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2024-25_q32_MS_1.png", "page": 5, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 550.0, 550.0, 721.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

31. SQP_2024-25_q36_A

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 36 A

Original section: Section-D

Source marks (nonadditive): 5

Parent question label: 36

Alternative: 36 A

Choice group: 2024-25_Q36

Exam mark group: 2024-25_Q36

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [12, 13]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/abaede5c255dc430e270.png", "assets/research/78a532d327a01badfb7a.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2024-25_q36_A_MS_1.png", "page": 8, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 634.0, 550.0, 756.0]}, {"file": "research/sqp/evidence/recent_2024-25_q36_A_MS_2.png", "page": 9, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 53, 550.0, 285.2]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

32. SQP_2024-25_q36_B

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 36 B

Original section: Section-D

Source marks (nonadditive): 5

Parent question label: 36

Alternative: 36 B

Choice group: 2024-25_Q36

Exam mark group: 2024-25_Q36

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [13]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/96d9bbec2f4cbdf9b005.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2024-25_q36_B_MS_1.png", "page": 9, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 299, 550.0, 575]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

33. SQP_2024-25_q36_VI_A

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 36 VI A

Original section: Section-D

Source marks (nonadditive): 5

Parent question label: 36

Alternative: 36 VI A

Choice group: 2024-25_Q36

Exam mark group: 2024-25_Q36

Marks include choice: True

Source verified: True

Answer verified: False

Source PDF page(s), one-based: [13]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: source_ready_answer_hold

notes: Official MS omits the factor of five fans. Its Rs. 546.00 answer is retained. Correct calculation from SQP: $(5 \times 0.1 \times 4 + 0.5 \times 2) \times 60 = 180 \text{ kWh}$; cost Rs.1170. MS marks $2+3=5$. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official energy cost omits four fans; correct Rs1170 documented separately. Consult the linked original marking scheme with this limitation in mind.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/b5fca59691633807b21e.png"]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

34. SQP_2024-25_q36_VI_B

Source ID: src-e13b537c0a62

Source title: CBSE Class X Science (086) Sample Question Paper 2024-25 (English)

Exam type: SQP

Year: 2024-25

Session: 2024-25

Paper code: Not recorded

Set: Not recorded

Original question label: 36 VI B

Original section: Section-D

Source marks (nonadditive): 5

Parent question label: 36

Alternative: 36 VI B

Choice group: 2024-25_Q36

Exam mark group: 2024-25_Q36

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [13]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Total 5 marks from Q36 scheme heading; this alternative has no explicit subpart mark breakdown. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/a1c02b5217abe859ea2f.pdf

Source raster asset(s): ["assets/research/380082ca45a93c597f5d.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2024-25_q36_VI_B_MS_1.png", "page": 10, "pdf_file": "research/sqp/sources/2024-25_Science-MS.pdf", "rect": [45.0, 140.7, 550.0, 489]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

35. SQP_2025-26_q34_A

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 34 A

Original section: Section – C

Source marks (nonadditive): 2

Parent question label: 34

Alternative: 34 A

Choice group: 2025-26_Q34

Exam mark group: 2025-26_Q34

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [11]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/0f61920aef4cdda16fc8.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q34_A_MS_1.png", "page": 7, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 233, 550.0, 369]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

36. SQP_2025-26_q34_B

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 34 B

Original section: Section - C

Source marks (nonadditive): 2

Parent question label: 34

Alternative: 34 B

Choice group: 2025-26_Q34

Exam mark group: 2025-26_Q34

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [12]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer

evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/109e842346e302071244.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q34_B_MS_1.png", "page": 7, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 369.0, 550.0, 537.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

37. SQP_2025-26_q34_VI_A

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 34 VI A

Original section: Section – C

Source marks (nonadditive): 2

Parent question label: 34

Alternative: 34 VI A

Choice group: 2025-26_Q34

Exam mark group: 2025-26_Q34

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [12]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/5f403b06ec568c3905ac.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q34_VI_A_MS_1.png", "page": 7, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 558.0, 550.0, 741.0]}, {"file": "research/sqp/evidence/recent_2025-26_q34_VI_A_MS_2.png", "page": 8, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 53.0, 550.0, 143.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

38. SQP_2025-26_q34_VI_B

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 34 VI B

Original section: Section – C

Source marks (nonadditive): 2

Parent question label: 34

Alternative: 34 VI B

Choice group: 2025-26_Q34

Exam mark group: 2025-26_Q34

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [12]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/5d76edf8f95ed15e0a47.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q34_VI_B_MS_1.png", "page": 8, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 149.0, 550.0, 500.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

39. SQP_2025-26_q36

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 36

Original section: Section – C

Source marks (nonadditive): 3

Parent question label: 36

Alternative: 36

Choice group: 2025-26_Q36

Exam mark group: 2025-26_Q36

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [13]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Official scheme rounds area to $5 \times 10^{-7} \text{ m}^2$ before calculating 3.33 m. Original “are” typo retained. No subpart marks printed. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/4c80e64970a4ee9819cd.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q36_MS_1.png", "page": 9, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 223.0, 550.0, 738.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

40. SQP_2025-26_q39_A

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 39 A

Original section: Section – C

Source marks (nonadditive): 5

Parent question label: 39

Alternative: 39 A

Choice group: 2025-26_Q39

Exam mark group: 2025-26_Q39

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [14]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Original “battery.,” retained. Total 5 marks; MS provides no subpart allocation. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/388ca2ab2430aa0d3e99.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q39_A_MS_1.png", "page": 12, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 629.0, 550.0, 744.0]}, {"file": "research/sqp/evidence/recent_2025-26_q39_A_MS_2.png", "page": 13, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 68, 550.0, 205]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

41. SQP_2025-26_q39_B

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 39 B

Original section: Section – C

Source marks (nonadditive): 5

Parent question label: 39

Alternative: 39 B

Choice group: 2025-26_Q39

Exam mark group: 2025-26_Q39

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [15]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Total 5 marks; MS provides no subpart allocation. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/16d6e0d4f1ad20ddc45c.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q39_B_MS_1.png", "page": 13, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 205, 550.0, 517.9]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

42. SQP_2025-26_q39_VI_A

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 39 VI A

Original section: Section – C

Source marks (nonadditive): 5

Parent question label: 39

Alternative: 39 VI A

Choice group: 2025-26_Q39

Exam mark group: 2025-26_Q39

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [15]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Full alternative is included: fuse series placement and heating protection are explicitly NCERT Electricity section 11.7.1 (printed p190); appliance parallel connections are section 11.6. This overlaps domestic-circuit applications but requires no magnetic effects. Full printed total is 5; MS does not specify subpart allocations. Original source typos retained. Answer text is source transcription; source crops are authoritative.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/2912cda893d3112877ba.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q39_VI_A_MS_1.png", "page": 13, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 531.5, 550.0, 744]}, {"file": "research/sqp/evidence/recent_2025-26_q39_VI_A_MS_2.png", "page": 14, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 53, 550.0, 527.1]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

43. SQP_2025-26_q39_VI_B

Source ID: src-f732f697f166

Source title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

Exam type: SQP

Year: 2025-26

Session: 2025-26

Paper code: Not recorded

Set: Not recorded

Original question label: 39 VI B

Original section: Section – C

Source marks (nonadditive): 5

Parent question label: 39

Alternative: 39 VI B

Choice group: 2025-26_Q39

Exam mark group: 2025-26_Q39

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [15]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Total 5 marks from scheme; no subpart mark allocation printed. Original “Heat generate” and “both resistor” retained. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/ec9f01d34a0fcf3ae035.pdf

Source raster asset(s): ["assets/research/846561eeb372e3828e4a.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2025-26_q39_VI_B_MS_1.png", "page": 14, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 541.0, 550.0, 742.0]}, {"file": "research/sqp/evidence/recent_2025-26_q39_VI_B_MS_2.png", "page": 15, "pdf_file": "research/sqp/sources/2025-26_Science-MS.pdf", "rect": [45.0, 53.0, 550.0, 566.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

44. SQP_2026-27_q31

Source ID: src-5ed9b8335767

Source title: CBSE Class X Science (086) Sample Question Paper 2026-27 (English)

Exam type: SQP

Year: 2026-27

Session: 2026-27

Paper code: Not recorded

Set: Not recorded

Original question label: 31

Original section: Section – C

Source marks (nonadditive): 1

Parent question label: 31

Alternative: 31

Choice group: 2026-27_Q31

Exam mark group: 2026-27_Q31

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [11]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/09bf85a26357d9b3c449.pdf

Source raster asset(s): ["assets/research/78aa2ffbaf3bc4ced1f2.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2026-27_q31_MS_1.png", "page": 7, "pdf_file": "research/sqp/sources/2026-27_Science-MS.pdf", "rect": [45.0, 125.0, 550.0, 147.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

45. SQP_2026-27_q34_A

Source ID: src-5ed9b8335767

Source title: CBSE Class X Science (086) Sample Question Paper 2026-27 (English)

Exam type: SQP

Year: 2026-27

Session: 2026-27

Paper code: Not recorded

Set: Not recorded

Original question label: 34 A

Original section: Section – C

Source marks (nonadditive): 2

Parent question label: 34

Alternative: 34 A

Choice group: 2026-27_Q34

Exam mark group: 2026-27_Q34

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [12]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/09bf85a26357d9b3c449.pdf

Source raster asset(s): ["assets/research/5f48365fa3a31499fddc.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2026-27_q34_A_MS_1.png", "page": 7, "pdf_file": "research/sqp/sources/2026-27_Science-MS.pdf", "rect": [45.0, 321.0, 550.0, 555.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

46. SQP_2026-27_q34_B

Source ID: src-5ed9b8335767

Source title: CBSE Class X Science (086) Sample Question Paper 2026-27 (English)

Exam type: SQP

Year: 2026-27

Session: 2026-27

Paper code: Not recorded

Set: Not recorded

Original question label: 34 B

Original section: Section – C

Source marks (nonadditive): 2

Parent question label: 34

Alternative: 34 B

Choice group: 2026-27_Q34

Exam mark group: 2026-27_Q34

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [12]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Original MS "thick-nesses" retained; comparison implicitly at fixed temperature. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/09bf85a26357d9b3c449.pdf

Source raster asset(s): ["assets/research/d235df1ef7aee4553a31.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2026-27_q34_B_MS_1.png", "page": 7, "pdf_file": "research/sqp/sources/2026-27_Science-MS.pdf", "rect": [45.0, 575.0, 550.0, 743.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

47. SQP_2026-27_q37_main

Source ID: src-5ed9b8335767

Source title: CBSE Class X Science (086) Sample Question Paper 2026-27 (English)

Exam type: SQP

Year: 2026-27

Session: 2026-27

Paper code: Not recorded

Set: Not recorded

Original question label: 37 main

Original section: Section – C

Source marks (nonadditive): 3

Parent question label: 37

Alternative: 37 main

Choice group: 2026-27_Q37

Exam mark group: 2026-27_Q37

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [13]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Original and reduced circuit diagrams are preserved in answer evidence. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/09bf85a26357d9b3c449.pdf

Source raster asset(s): ["assets/research/be21a00c46d5bfcd70a0.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2026-27_q37_main_MS_1.png", "page": 9, "pdf_file": "research/sqp/sources/2026-27_Science-MS.pdf", "rect": [45.0, 53, 550.0, 663]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

48. SQP_2026-27_q37_VI

Source ID: src-5ed9b8335767

Source title: CBSE Class X Science (086) Sample Question Paper 2026-27 (English)

Exam type: SQP

Year: 2026-27

Session: 2026-27

Paper code: Not recorded

Set: Not recorded

Original question label: 37 VI

Original section: Section – C

Source marks (nonadditive): 3

Parent question label: 37

Alternative: 37 VI

Choice group: 2026-27_Q37

Exam mark group: 2026-27_Q37

Marks include choice: True

Source verified: True

Answer verified: True

Source PDF page(s), one-based: [13]

choice_resolved: True

verification_scope: Question wording and provenance checked against official source; verified does not imply official answer is error-free.

publication_status: ready

notes: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

pdf_file: assets/research/09bf85a26357d9b3c449.pdf

Source raster asset(s): ["assets/research/e375be7307a6f48459ee.png"]

answer_transcription: Source transcription/summary; source MS crops are authoritative. See notes for source errors and any explicitly independently derived portions.

answer_crop_regions: [{"file": "research/sqp/evidence/recent_2026-27_q37_VI_MS_1.png", "page": 9, "pdf_file": "research/sqp/sources/2026-27_Science-MS.pdf", "rect": [45.0, 663.5, 550.0, 743]}, {"file": "research/sqp/evidence/recent_2026-27_q37_VI_MS_2.png", "page": 10, "pdf_file": "research/sqp/sources/2026-27_Science-MS.pdf", "rect": [45.0, 53.0, 550.0, 112.0]}]

Official source: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Official marking scheme: https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf